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1. Introduction

This document models connected coffee machines that send `telemetry` to a `central cloud site`, receive OTA firmware updates, and produce operational and audit logging `evidence`. Authored functional architecture remains stable while runtime, code, and verification layers are inferred from infrastructure, source, and test artifacts.

1.1. Scope and Assumptions

- `Functional Groups` and `Functional Units` are the stable authored design backbone.
- `Runtime elements` and `code elements` are inferred from available IaC, deployment, and source artifacts.
- Inferred coverage is evidence-driven and may be partial when source metadata is missing.
- `Requirements` and reference indexes are provided for traceability, not as primary narrative flow.

2. How To Read This Document

This document is intentionally split into two layers:

1. Concept layer: functional and view-specific architecture narratives for human understanding.
2. Evidence layer: generated requirement traceability and inferred reference indexes.

Read in this order:

1. Architecture Intent View: stable authored architecture intent and ownership boundaries.
2. Communication / Deployment / Security Views: viewpoint-specific operational behavior.
3. Traceability View: requirement mapping, evidence coverage, and confidence.
4. Reference Appendix: supporting reference IDs and inferred-source links.

3. Document Health Snapshot

View	Authored Status	Authored Status Explanation	Units in Scope	Direct Evidence Coverage	Gap Count	Heuristic Confidence	Heuristic Basis
Architecture Intent View (`VIEW-ARCHITECTURE-INTENT`)	stable	Stable authored capability boundaries and responsibility ownership.	7	7	0	high	Defaults authored units to covered; use authoredStatus as the primary publication signal.
Communication View (`VIEW-COMMUNICATION`)	in-review	Communication handoffs are mapped; protocol detail remains partially inferred.	7	4	3	medium	Counts a unit as covered when direct inferred runtime evidence exists.
Deployment View (`VIEW-DEPLOYMENT`)	in-review	Deployment/runtime mapping is present with evolving control-plane evidence depth.	7	4	3	medium	Counts a unit as covered when direct inferred runtime evidence exists.
Security View (`VIEW-SECURITY`)	draft	Core threats are modeled while additional control evidence is being expanded.	7	7	1	high	Counts a unit as covered when explicit authored attack-vector mapping exists.
Traceability View (`VIEW-TRACEABILITY`)	in-review	Requirement mappings exist; verification and evidence coverage continue to be hardened.	7	7	3	high	Counts a unit as covered when direct inferred runtime or code evidence exists.
State Lifecycle View (`VIEW-STATE-LIFECYCLE`)	in-review	OTA lifecycle transitions and guards are authored for deterministic operational progression.	0	0	0	n/a	Lifecycle heuristic is currently basic; treat authoredStatus as primary signal.
Interaction Flow View (`VIEW-INTERACTION-FLOW`)	in-review	OTA rollout interaction sequence is explicitly modeled from operator intent to rollout reporting.	5	5	0	high	Flow heuristic is step/branch-oriented; treat authoredStatus plus projection completeness as the primary signal.

4. Terms and Definitions

Term	Definition
actor (TERM-ACTOR)	A person or operational role that interacts with functional units.
architecture decision (TERM-DECISION)	Authored architecture decision record with status, context, decision text, and consequences.
AsciiDoc diagram (TERM-ASCIIDOC-DIAGRAM)	Generated diagram block, usually Mermaid, that visualizes architecture relationships in the AsciiDoc publication.
AsciiDoc document (TERM-ASCIIDOC-DOCUMENT)	Human-readable generated architecture publication document assembled from authored model, inferred evidence, diagrams, references, and decisions.
AsciiDoc section (TERM-ASCIIDOC-SECTION)	Structured generated document section or row model used to render authored and inferred architecture content.
attack vector (TERM-ATTACK-VECTOR)	A technical misuse or attack path that targets functional, referenced, or runtime elements.
audit log record (DATA-AUDIT-LOG-RECORD)	Immutable log record for security and compliance analysis. Aliases:audit records, logging records
audit logging is enabled (FEAT-AUDIT-LOGGING)	Feature flag enabling audit-grade log persistence. Aliases:audit enabled
authored mapping (TERM-AUTHORED-MAPPING)	An explicit relationship declared in architecture inputs between authored or referenced elements.
barista (ACT-BARISTA)	Operates coffee machines and observes local machine health. Aliases:store operator
brew cycle completed event is received (EVT-BREW-CYCLE-COMPLETED)	Event raised when a coffee machine completes a brew cycle. Aliases:brew completed, brew cycle event
brew telemetry (DATA-BREW-TELEMETRY)	Machine telemetry including brew duration, temperature, and status metadata. Aliases:machine status, machine telemetry, telemetry, telemetry payloads, telemetry records
campaign id (DATA-CAMPAIGN-ID)	Stable OTA campaign identifier used to correlate rollout and audit records. Aliases:campaign identifier, campaign ids
catalog (TERM-CATALOG)	Controlled vocabulary document used to normalize model, requirement, and generated-document terminology.
catalog entry (TERM-CATALOG-ENTRY)	A stable catalog term with a definition and aliases for linting, linking, and generated references.
CLI command (TERM-CLI-COMMAND)	Command-line entrypoint that coordinates loading, validation, generation, and file output for an engineering-model workflow.
cloud control (FG-CLOUD-CONTROL)	Capability area for fleet ingestion, campaign orchestration, and cloud reporting. Aliases:central cloud site

Term	Definition
cloud ingest endpoint is unavailable (EVT-CLOUD-INGEST-ENDPOINT-UNAVAILABLE)	Event raised when cloud ingest endpoint becomes unavailable for telemetry ingestion. Aliases:endpoint outage, ingest endpoint unavailable
cloud logging service (REF-CLOUD-LOGGING-SERVICE)	Managed log and metrics service used for fleet observability. Aliases:logging service
cloud runtime operations (FU-CLOUD-RUNTIME-OPERATIONS)	Operates cloud runtime deployment and scaling controls. Aliases:runtime operations
cloud runtime sdk (REF-CLOUD-RUNTIME-SDK)	SDK used by cloud units for runtime integrations. Aliases:cloud sdk
code element (TERM-CODE-ELEMENT)	An inferred code structure or ownership element discovered from source trees and build metadata.
coffee fleet system (SYS-COFFEE-FLEET-SYSTEM)	Connected coffee-machine fleet and central cloud control system-of-interest covered by this architecture. Aliases:the coffee fleet system
coffee machine fleet operation is active (STATE-FLEET-OPERATION-ACTIVE)	State where machines and central cloud are actively serving operations. Aliases:fleet operation active
compliance mapping (TERM-COMPLIANCE-MAPPING)	Authored mapping that links a model control to selected compliance controls, model scope, implementation status, evidence, and responsible roles.
control (TERM-CONTROL)	An authored security, policy, or operational control used to constrain behavior and reduce risk.
control verification (TERM-CONTROL-VERIFICATION)	Authored control verification record with method, status, threat/risk scope, findings, and evidence.
data object (TERM-DATA-OBJECT)	An authored data artifact or contract shape traced across flow, interface, and implementation boundaries.
deployment target (TERM-DEPLOYMENT-TARGET)	An authored deployment destination such as environment, cluster, namespace, or registry scope.
design document (TERM-DESIGN)	Authored design narrative organized by model entities and architecture views.
design view (TERM-DESIGN-VIEW)	View-scoped design narrative attached to an authored functional group or functional unit.
device identity and secrets (FU-DEVICE-IDENTITY-SECRETS)	Manages device identity material and secure configuration. Aliases:device secrets
event (TERM-EVENT)	An authored trigger signal that drives transitions, flow progress, or asynchronous outcomes.
evidence (TERM-EVIDENCE)	A file, artifact, or observation used to support control, risk, threat, or verification claims.
firmware bundle store (REF-FIRMWARE-BUNDLE-STORE)	Object store containing signed OTA firmware bundles. Aliases:firmware store

Term	Definition
`firmware signature is valid` (COND-FIRMWARE-SIGNATURE-VALID)	Condition indicating firmware signature verification succeeded. Aliases:signature valid
`firmware signer service` (REF-FIRMWARE-SIGNER-SERVICE)	Service that signs firmware artifacts before OTA rollout. Aliases:signer service
`firmware validation failed event is received` (EVT-FIRMWARE-VALIDATION-FAILED)	Event raised when signature/hash validation fails on firmware. Aliases:validation failed
`fleet ingestion api` (FU-FLEET-INGESTION-API)	Ingests telemetry and machine status from edge devices. Aliases:ingestion api
`fleet observability reporting` (FU-FLEET-OBSERVABILITY-REPORTING)	Aggregates telemetry and update outcomes into operational and audit signals. Aliases:observability reporting
`fleet operator` (ACT-FLEET-OPERATOR)	Owns OTA rollout campaigns and fleet-level reliability. Aliases:fleet admin
`flow` (TERM-FLOW)	An authored causal interaction sequence from user/system intent to outcome, represented as ordered steps.
`flow step` (TERM-FLOW-STEP)	A single authored step in a flow that captures action, data in/out, references, and normal/error/async transitions.
`functional group` (TERM-FUNCTIONAL-GROUP)	A major authored capability area that groups related functional units.
`functional unit` (TERM-FUNCTIONAL-UNIT)	An authored working unit inside a functional group that owns specific behavior.
`generation workflow` (TERM-GENERATION-WORKFLOW)	Orchestrated run that combines model loading, validation, projection, rendering, and artifact emission.
`inference hint` (TERM-INFERENCE-HINT)	Authored source and ownership configuration used to discover runtime, code, and verification evidence.
`inferred mapping` (TERM-INFERRED-MAPPING)	A discovered relationship that links inferred runtime/code elements upward to authored design.
`interface` (TERM-INTERFACE)	An authored interface boundary (for example API, channel, endpoint, or contract) owned by a functional unit.
`iot ingest endpoint` (REF-IOT-INGEST-ENDPOINT)	Cloud endpoint for receiving telemetry and status from devices. Aliases:iot endpoint
`lint run` (TERM-LINT-RUN)	Requirement lint configuration that controls parsing and quality checks for requirement text.
`LOBSTER activity trace` (TERM-LOBSTER-ACTIVITY-TRACE)	Generated LOBSTER activity JSON that links verification evidence back to formal requirement identifiers.

Term	Definition
`log tampering` (AV-LOG-TAMPERING)	Attempts to suppress or alter security/operational log records. Aliases:audit tampering
`machine edge` (FG-MACHINE-EDGE)	Capability area for device-local telemetry and OTA execution behavior. Aliases:coffee machine edge
`machine identity` (DATA-MACHINE-IDENTITY)	Device identity material used for authenticated publish, ingest, and audit attribution. Aliases:device identity, machine identities
`machine telemetry collection` (FU-MACHINE-TELEMETRY-COLLECTION)	Collects brew/device telemetry and publishes cloud-bound payloads. Aliases:telemetry collection
`malicious firmware bundle` (AV-MALICIOUS-FIRMWARE-BUNDLE)	Tampered firmware payload attempts to execute on devices. Aliases:firmware tampering
`MCP server` (TERM-MCP-SERVER)	Model Context Protocol server that exposes engineering-model operations to AI agents through JSON-RPC tools.
`MCP stdio transport` (TERM-MCP-STDIO-TRANSPORT)	Content-Length framed standard input/output transport used by the MCP command-line server.
`MCP tool` (TERM-MCP-TOOL)	Named MCP operation with a constrained input schema and structured response payload.
`MCP tool response` (TERM-MCP-TOOL-RESPONSE)	Structured MCP tool payload that includes success state, schema version, generated timestamp, and result data or validation error.
`model` (TERM-MODEL)	The authored architecture model root that composes functional structure, relationships, inference hints, and views.
`model bundle` (TERM-BUNDLE)	Loaded architecture, catalog, and companion documents resolved as one working model context.
`mqtt device sdk` (REF-MQTT-DEVICE-SDK)	SDK used by edge components for telemetry and command transport. Aliases:mqtt sdk
`normal mode is active` (MODE-NORMAL)	Default operation mode for telemetry and OTA behavior. Aliases:normal mode, normal operation mode
`Open OTM document` (TERM-OPEN-OTM-DOCUMENT)	Open Threat Model JSON representation generated from the engineering model for interoperable threat-model exchange.
`OSCAL assessment results` (TERM-OSCAL-ASSESSMENT-RESULTS)	Generated OSCAL assessment results that summarize reviewed controls, verification findings, and modeled risks.
`OSCAL POA&M` (TERM-OSCAL-POAM)	Generated OSCAL plan of action and milestones that maps modeled POA&M items and related risks into compliance JSON.
`OSCAL SSP` (TERM-OSCAL-SSP)	Generated OSCAL system security plan that maps authored system metadata, components, controls, allocations, and evidence into SSP JSON.
`ota campaign is enabled` (FEAT-OTA-CAMPAIGN)	Feature flag enabling centralized OTA rollout behavior. Aliases:ota enabled, ota rollout

Term	Definition
`ota campaign policy` (DATA-OTA-CAMPAIGN-POLICY)	Policy describing rollout cohorts, windows, and rollback criteria. Aliases:campaign policy, rollout cohorts, rollout windows
`ota campaign scheduled event is received` (EVT-OTA-CAMPAIGN-SCHEDULED)	Event raised when a fleet OTA campaign is scheduled. Aliases:campaign commands, campaign scheduled, ota commands, ota rollout commands
`ota status` (DATA-OTA-STATUS)	Device-level status for firmware download, apply, and rollback outcomes. Aliases:firmware apply, ota outcomes, rollback firmware apply, update outcomes, update/rollback status
`ota status reported event is received` (EVT-OTA-STATUS-REPORTED)	Event raised when device reports OTA apply status. Aliases:apply status, apply/rollback status, ota status reported, rollback status, status feedback, update status
`ota update agent` (FU-OTA-UPDATE-AGENT)	Downloads, validates, and applies signed firmware updates on machines. Aliases:device update agent
`platform operations` (FG-PLATFORM-OPERATIONS)	Capability area for runtime operations and device identity/secrets lifecycle. Aliases:fleet platform operations
`platform operator` (ACT-PLATFORM-OPERATOR)	Owns runtime deployment and operational controls. Aliases:platform ops
`POA&M item` (TERM-POAM-ITEM)	Plan of action and milestones item linked to a modeled risk and supporting artifacts.
`reference index` (TERM-REFERENCE-INDEX)	Generated index of authored, catalog, runtime, code, and verification references with stable anchors and backlinks.
`referenced element` (TERM-REFERENCED-ELEMENT)	An architecture-relevant external, platform, or third-party dependency represented by role.
`repository index` (TERM-REPO-INDEX)	Bounded first-party source index used to answer model-aware file, requirement, control, and threat lookup queries.
`requirement` (TERM-REQUIREMENT)	A structured requirement statement that defines expected system behavior and maps to functional ownership.
`risk` (TERM-RISK)	Authored risk record with likelihood, impact, response, scope, controls, and evidence.
`runtime element` (TERM-RUNTIME-ELEMENT)	An inferred runtime realization element discovered from infrastructure and deployment sources.
`security analyst` (ACT-SECURITY-ANALYST)	Reviews identity, firmware integrity, and logging controls. Aliases:security reviewer
`security context` (TERM-SECURITY-CONTEXT)	Security-focused generated context view that groups owned functional units and shows external actors, references, flows, controls, and trust boundaries.
`signed firmware bundle` (DATA-FIRMWARE-BUNDLE)	Signed OTA firmware artifact distributed to machine cohorts for update apply. Aliases:firmware artifact, firmware bundle, firmware bundles, signed firmware bundles
`source block` (TERM-SOURCE-BLOCK)	Stable source reference block that records file path, optional line span, kind, summary, and linked entity IDs.

Term	Definition
spoofed device identity (AV-SPOOFED-DEVICE-IDENTITY)	Unauthorized actor impersonates a machine identity. Aliases:device spoofing
state (TERM-STATE)	An authored lifecycle state used to model transition behavior, guards, and event-driven progression.
Structurizr DSL (TERM-STRUCTURIZR-DSL)	Generated Structurizr DSL workspace that projects authored architecture, relationships, dynamic views, and deployment metadata.
telemetry ingested event is received (EVT-TELEMETRY-INGESTED)	Event raised when cloud ingestion accepts telemetry. Aliases:fleet events, ingestion events, observability event fanout, observability events, telemetry ingested
telemetry replay abuse (AV-TELEMETRY-REPLAY-ABUSE)	Replayed telemetry floods ingestion and distorts reporting. Aliases:telemetry replay
telemetry replay abuse is detected (EVT-TELEMETRY-REPLAY-ABUSE-DETECTED)	Event raised when replayed telemetry behavior is detected and flagged as abuse. Aliases:replay abuse, telemetry replay detected
threat assumption (TERM-THREAT-ASSUMPTION)	Authored security assumption that records scope, rationale, owner, status, and evidence.
Threat Dragon document (TERM-THREAT-DRAGON-DOCUMENT)	Threat Dragon JSON representation generated from the engineering model for STRIDE threat-model review.
threat mitigation (TERM-THREAT-MITIGATION)	Authored mitigation record that links a threat scenario to a control, effectiveness, verification, and evidence.
threat model export (TERM-THREAT-MODEL-EXPORT)	Generated security artifact that translates authored architecture, flows, trust boundaries, threats, and mitigations into an external threat-model schema.
threat out-of-scope decision (TERM-THREAT-OUT-OF-SCOPE)	Authored decision that excludes a threat concern from current scope with reason, owner, expiry, and evidence.
threat scenario (TERM-THREAT-SCENARIO)	Authored threat narrative that connects attack vectors, scope, flows, controls, risks, and verification evidence.
trace marker (TERM-TRACE-MARKER)	Source-level marker such as TRLC-LINKS or ENGMODEL-LINKS used to connect declarations to requirements and model entities.
TRLC package (TERM-TRLC-PACKAGE)	Generated TRLC model and requirements package used for formal requirement traceability processing.
trust boundary (TERM-TRUST-BOUNDARY)	An authored trust separation zone that marks policy, identity, or security control boundaries.
update campaign orchestration (FU-UPDATE-CAMPAIGN-ORCHESTRATION)	Selects target machine groups and coordinates OTA rollout phases. Aliases:campaign orchestration
upward linking (TERM-UPWARD-LINKING)	Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs.

Term	Definition
`validation` (TERM-VALIDATION)	Model quality gate that checks authored documents, references, relationship semantics, and requirement lint results.
`validation diagnostic` (TERM-VALIDATION-DIAGNOSTIC)	Structured validation finding with code, severity, message, and optional source path.
`verification check` (TERM-VERIFICATION-CHECK)	An inferred or authored verification artifact that validates requirement behavior with test evidence.
`view` (TERM-VIEW)	Authored projection configuration that selects roots, entity kinds, mappings, audience, and abstraction.

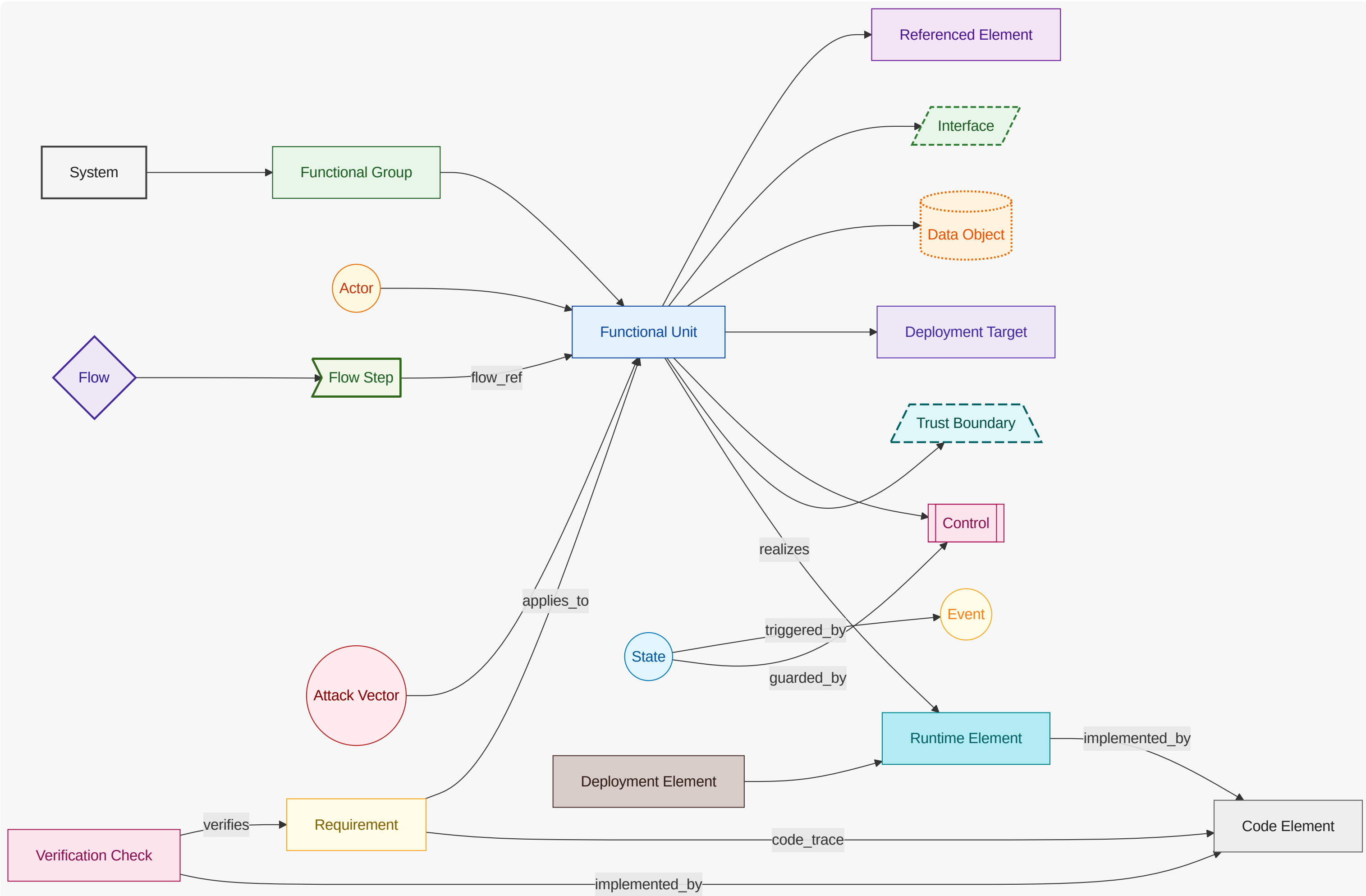
5. Architecture Decision Records

Architecture decision records are generated separately and linked here as a compact index.

ID	Title	Status	Date
ADR-COF-001	Separate edge OTA execution from cloud campaign orchestration	accepted	2026-04-29

5.1. Diagram Color and Shape Legend

This legend applies to all Mermaid diagrams in this document.



6. Architecture Intent View

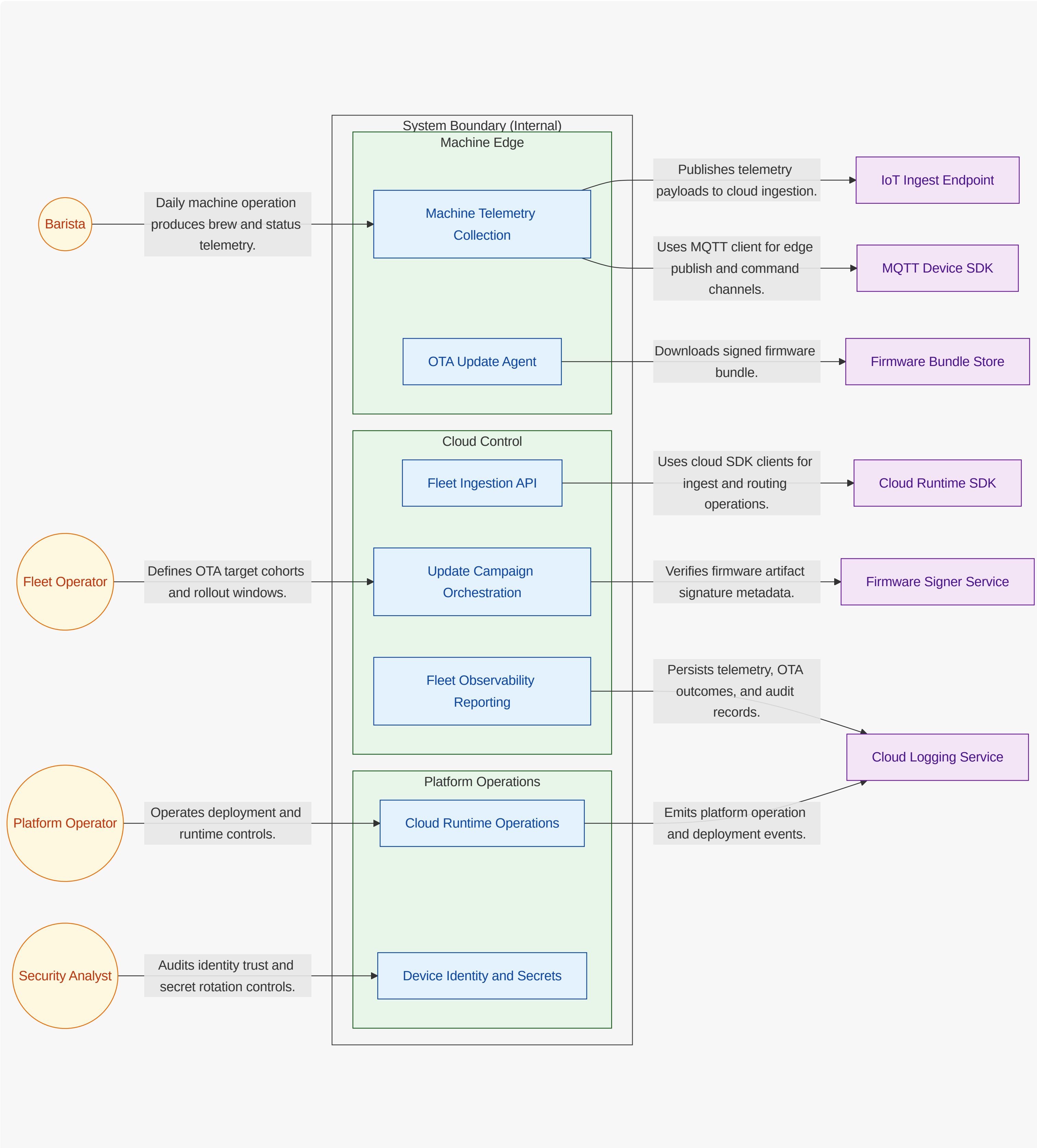
Show the stable authored architecture intent: capability areas, unit ownership boundaries, and intentional responsibility split.

6.1. What This View Answers

- What are the main capability areas?
- What are the stable units of responsibility and what does each unit own?
- How are responsibilities intentionally split?

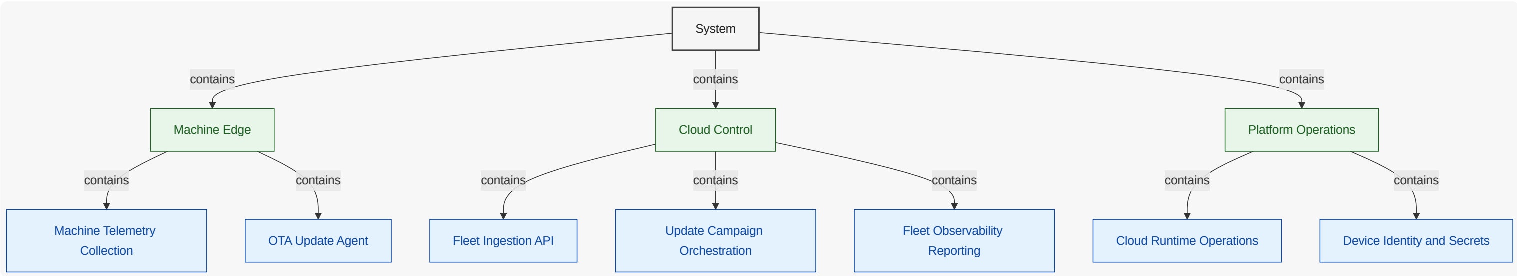
6.2. System Boundary Diagram

What this diagram shows: explicit internal vs external scope. Functional units are inside the `System Boundary (Internal)` container, while actors and referenced dependencies are shown outside the boundary.



6.3. System Decomposition Diagram

What this diagram shows: stable capability decomposition from functional groups into functional units.


System
contains
contains
contains
Machine Edge
Cloud Control
Platform Operations
contains
contains
contains
contains
contains
contains
contains
Machine Telemetry Collection
OTA Update Agent
Fleet Ingestion API
Update Campaign Orchestration
Fleet Observability Reporting
Cloud Runtime Operations
Device Identity and Secrets

6.4. Functional Group Manhattan Matrix

What this diagram shows: a table-like matrix where functional groups are the bottom row (columns) and functional units are arranged in rows above their owning group.

	Fleet Ingestion API	
Machine Telemetry Collection	Fleet Observability Reporting	Cloud Runtime Operations
OTA Update Agent	Update Campaign Orchestration	Device Identity and Secrets
Machine Edge	Cloud Control	Platform Operations

6.5. Functional Groups and Units (View Scoped)

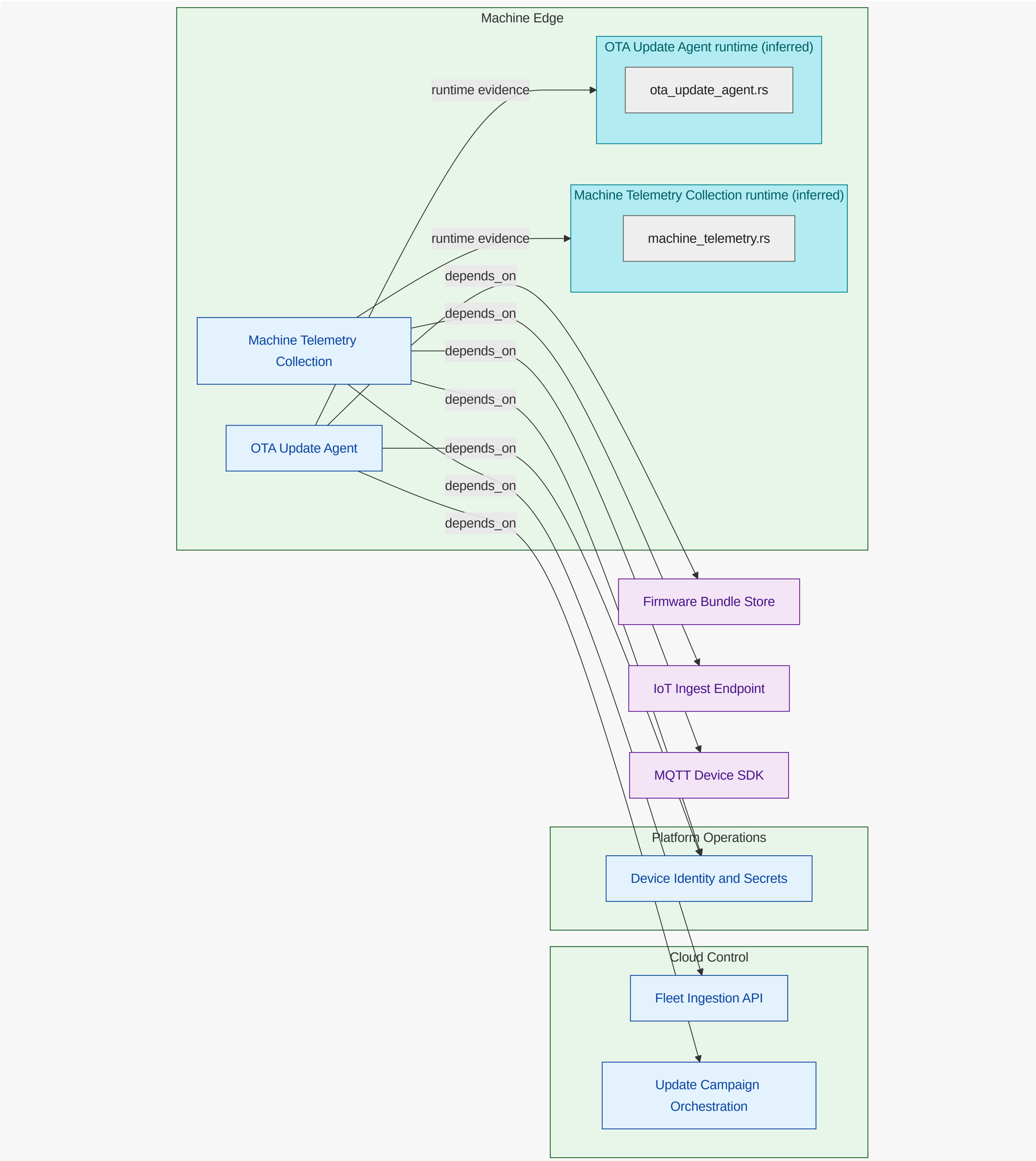
6.5.1. Machine Edge (FG-MACHINE-EDGE)

Device-local `telemetry` and OTA execution responsibilities.

```
    <<REF_IDX-CATALOG_FG_MACHINE_EDGE,Machine Edge>> owns behavior that executes directly on coffee machines.
It gathers <<REF_IDX-CATALOG_TELEMETRY,telemetry>>, applies OTA updates, and reports local status to cloud boundaries.
```

Functional Unit Dependency Diagram

What this diagram shows: dependencies involving functional units in this functional group. Units in this group are rendered inside the group container; connected external units and referenced elements are shown outside it.



Machine Telemetry Collection (FU-MACHINE-TELEMETRY-COLLECTION)

Collects brew/device `telemetry` and transmits cloud-bound `telemetry payloads`.

Owned Responsibility

functional responsibility for `machine telemetry collection`; includes decision and orchestration `flow` to `Device Identity and Secrets` and decision and orchestration `flow` to `Fleet Ingestion API`

Key Collaborators

`Device Identity and Secrets`, `Fleet Ingestion API`, `IoT Ingest Endpoint`, `MQTT Device SDK`

Linked Requirements

`REQ-COF-001`, `REQ-COF-006`, `REQ-COF-011`

OTA Update Agent (FU-OTA-UPDATE-AGENT)

Downloads `signed firmware bundles`, validates integrity, applies updates, and reports status.

Owned Responsibility

functional responsibility for `ota update agent`; includes decision and orchestration `flow` to `Device Identity and Secrets` and decision and orchestration `flow` to `Firmware Bundle Store`

Key Collaborators

`Device Identity and Secrets`, `Firmware Bundle Store`, `Update Campaign Orchestration`

Linked Requirements

`REQ-COF-003`, `REQ-COF-004`, `REQ-COF-005`, `REQ-COF-010`

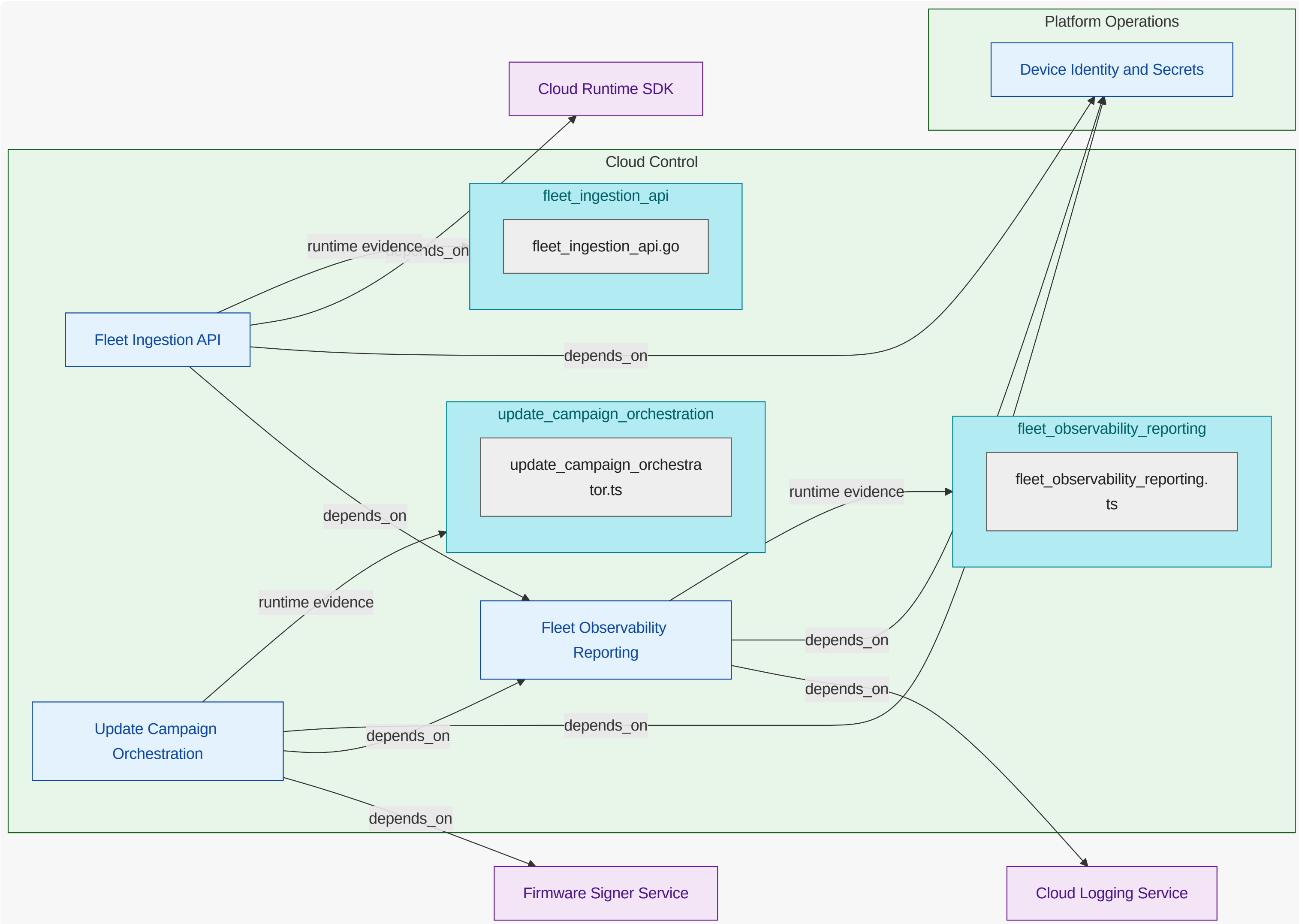
6.5.2. Cloud Control (FG-CLOUD-CONTROL)

Central cloud ingestion, campaign `control`, and fleet reporting.

```
    <<REF_IDX-CATALOG_FG_CLOUD_CONTROL,Cloud Control>> owns central APIs and coordination logic for fleet <<REF_IDX-CATALOG_TELEMETRY,telemetry>> and <<REF_IDX-CATALOG_OTA_ROLLOUT,OTA rollouts>>.
It provides global fleet visibility and deterministic rollout decisions.
```

Functional Unit Dependency Diagram

What this diagram shows: dependencies involving functional units in this functional group. Units in this group are rendered inside the group container; connected external units and referenced elements are shown outside it.



Fleet Ingestion API (FU-FLEET-INGESTION-API)

Receives `telemetry` and `machine status`, applies `validation`, and emits `ingestion events`.

Owned Responsibility

functional responsibility for `fleet ingestion api`; includes decision and orchestration `flow` to `Cloud Runtime SDK` and decision and orchestration `flow` to `Device Identity and Secrets`

Key Collaborators

`Cloud Runtime SDK`, `Device Identity and Secrets`, `Fleet Observability Reporting`

Linked Requirements

`REQ-COF-001`, `REQ-COF-002`, `REQ-COF-006`, `REQ-COF-008`, `REQ-COF-009`, `REQ-COF-011`, `REQ-COF-012`

Fleet Observability Reporting (FU-FLEET-OBSERVABILITY-REPORTING)

Aggregates `telemetry` and `OTA outcomes` into dashboards, alerts, and `audit records`.

Owned Responsibility

functional responsibility for `fleet observability reporting`; includes decision and orchestration `flow` to `Cloud Logging Service` and decision and orchestration `flow` to `Device Identity and Secrets`

Key Collaborators

`Cloud Logging Service`, `Device Identity and Secrets`

Linked Requirements

`REQ-COF-002`, `REQ-COF-005`, `REQ-COF-007`, `REQ-COF-008`, `REQ-COF-009`, `REQ-COF-010`, `REQ-COF-012`

Update Campaign Orchestration (FU-UPDATE-CAMPAIGN-ORCHESTRATION)

Plans OTA `rollout cohorts`, drives update commands, and handles rollback `flow` decisions.

Owned Responsibility

functional responsibility for `update campaign orchestration`; includes decision and orchestration `flow` to `Device Identity and Secrets` and decision and orchestration `flow` to `Firmware Signer Service`

Key Collaborators

`Device Identity and Secrets`, `Firmware Signer Service`, `Fleet Observability Reporting`

Linked Requirements

`REQ-COF-003`, `REQ-COF-005`, `REQ-COF-010`

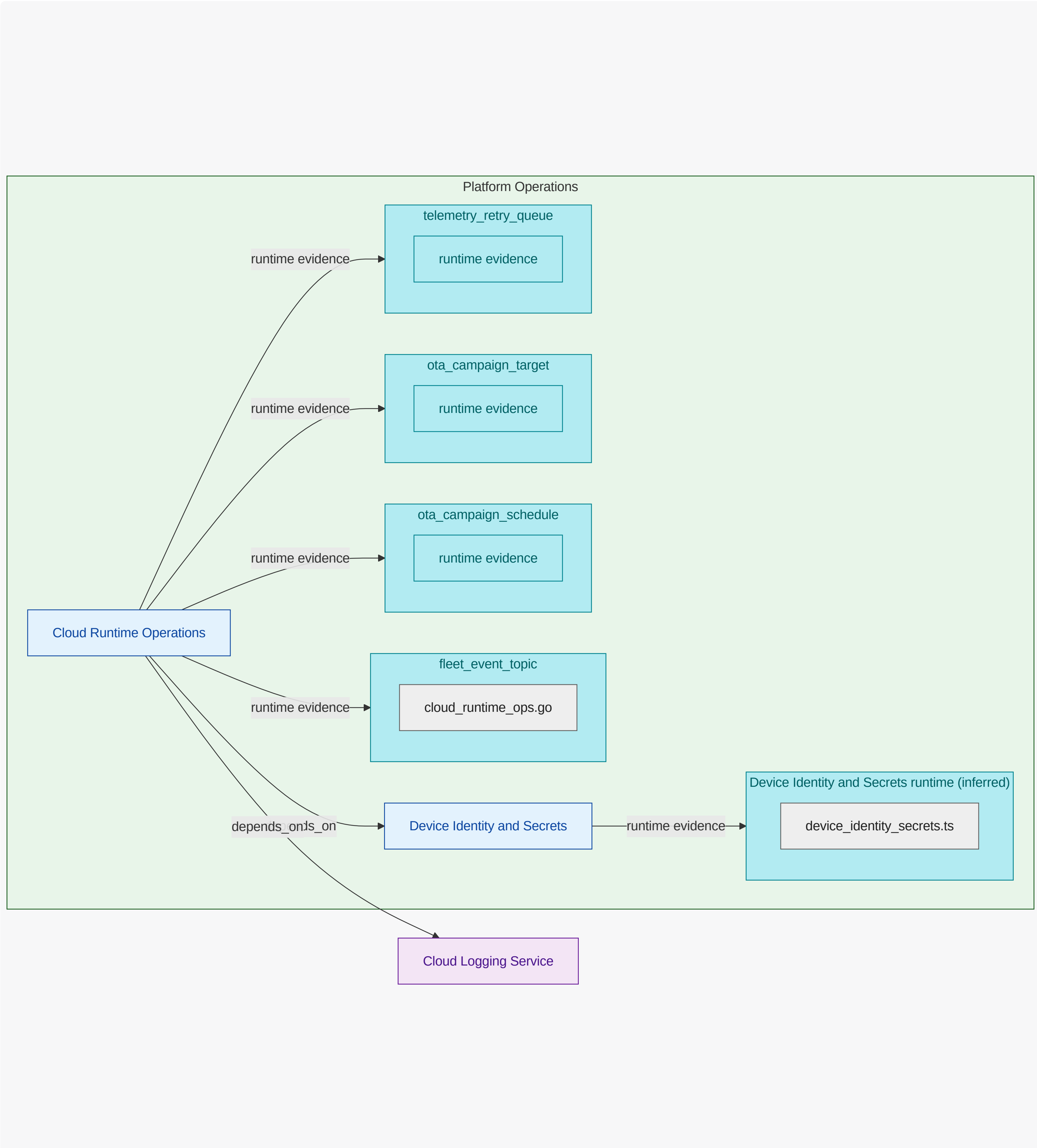
6.5.3. Platform Operations (FG-PLATFORM-OPERATIONS)

`Cloud runtime operations` plus identity and secret-management controls.

```
    <<REF_IDX-CATALOG_FG_PLATFORM_OPERATIONS,Platform Operations>> owns deployability, runtime guardrails, and secure
identity/config lifecycle.
It ensures cloud and device <<REF_IDX-ENGMODEL_TERM_TRUST_BOUNDARY,trust boundaries>> remain enforceable over time.
```

Functional Unit Dependency Diagram

What this diagram shows: dependencies involving functional units in this functional group. Units in this group are rendered inside the group container; connected external units and referenced elements are shown outside it.



Cloud Runtime Operations (FU-CLOUD-RUNTIME-OPERATIONS)

Owns runtime deployment lifecycle, scaling controls, and runtime operational safety.

Owned Responsibility

functional responsibility for `cloud runtime operations`; includes decision and orchestration `flow` to `Cloud Logging Service` and decision and orchestration `flow` to `Device Identity and Secrets`

Key Collaborators

`Cloud Logging Service`, `Device Identity and Secrets`

Linked Requirements

`REQ-COF-008`, `REQ-COF-012`

Device Identity and Secrets (FU-DEVICE-IDENTITY-SECRETS)

Manages device credentials, cloud tokens, and secure configuration used by edge/cloud units.

Owned Responsibility

functional responsibility for `device identity and secrets`

Key Collaborators

none

Linked Requirements

`REQ-COF-004`, `REQ-COF-007`

7. Communication View

Show who communicates with whom across key request paths, including `interface` edges and dependency handoffs.

7.1. What This View Answers

- Who talks to whom and over which interfaces?
- Which edges are synchronous, asynchronous, event-driven, or callback-oriented?
- Which communication paths are critical to the main request `flow`?

7.2. Coverage Gaps

Coverage summary: 4/7 units have direct `evidence` coverage in this `view`.

- `Device Identity and Secrets` has no direct inferred runtime/deployment `evidence` yet.
- `Machine Telemetry Collection` has no direct inferred runtime/deployment `evidence` yet.
- `OTA Update Agent` has no direct inferred runtime/deployment `evidence` yet.

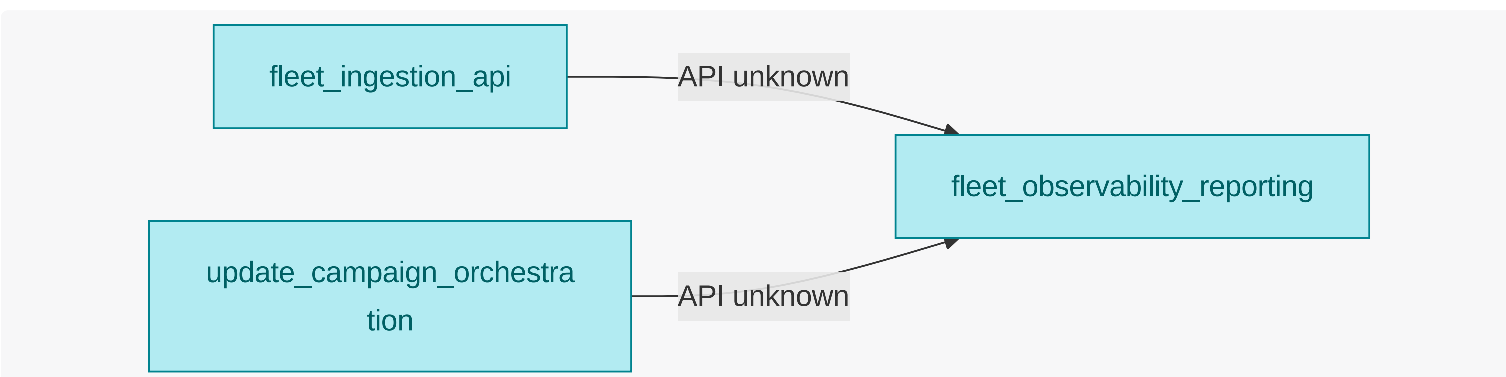
7.3. Recommended Next Evidence Additions

- Add explicit communication metadata (protocol, sync/async/`event`/callback) to `authored mappings` where currently implicit.
- Expose service protocol/port metadata in Helm or manifest values to enrich interface-edge inference.
- Capture missing communicating runtime units by adding discoverable workload artifacts under scanned inference roots.

7.4. Communication Path Graph

What this diagram shows: inferred unit-to-unit request/call flow, including available interface/port labels.

> **NOTE**
> This graph is inferred from available deployment/runtime artifacts and may be partial. Missing edges or interface details indicate missing derived communication evidence, not necessarily missing behavior.



7.5. Functional Groups and Units (View Scoped)

8. Deployment View

Show inferred deployment artifacts and ownership mapping to authored units. This `view` emphasizes deployment relationships and `platform operations`.

8.1. What This View Answers

- How does authored intent become deployed runtime?
- What control-plane objects manage deployable units?
- Where are platform ownership boundaries and missing ownership signals?

8.2. Coverage Gaps

Coverage summary: 4/7 units have direct `evidence` coverage in this `view`.

- `Device Identity and Secrets` has no direct inferred runtime/deployment `evidence` yet.
- `Machine Telemetry Collection` has no direct inferred runtime/deployment `evidence` yet.
- `OTA Update Agent` has no direct inferred runtime/deployment `evidence` yet.

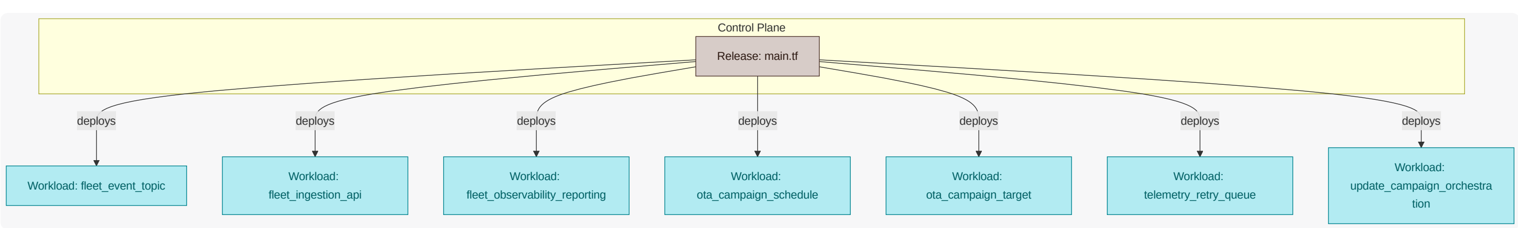
8.3. Recommended Next Evidence Additions

- Add owner metadata to Flux/Kustomization/Helm objects for clearer platform ownership mapping.
- Ensure deployment `control` artifacts (source, kustomization, release) are all included in scanned directories.
- Add deterministic naming conventions for release and namespace objects to improve cross-artifact linking.

8.4. Deployment Artifact Relationship Graph

What this diagram shows: how source/control-plane artifacts connect to releases, namespaces, workloads, and target cluster context.

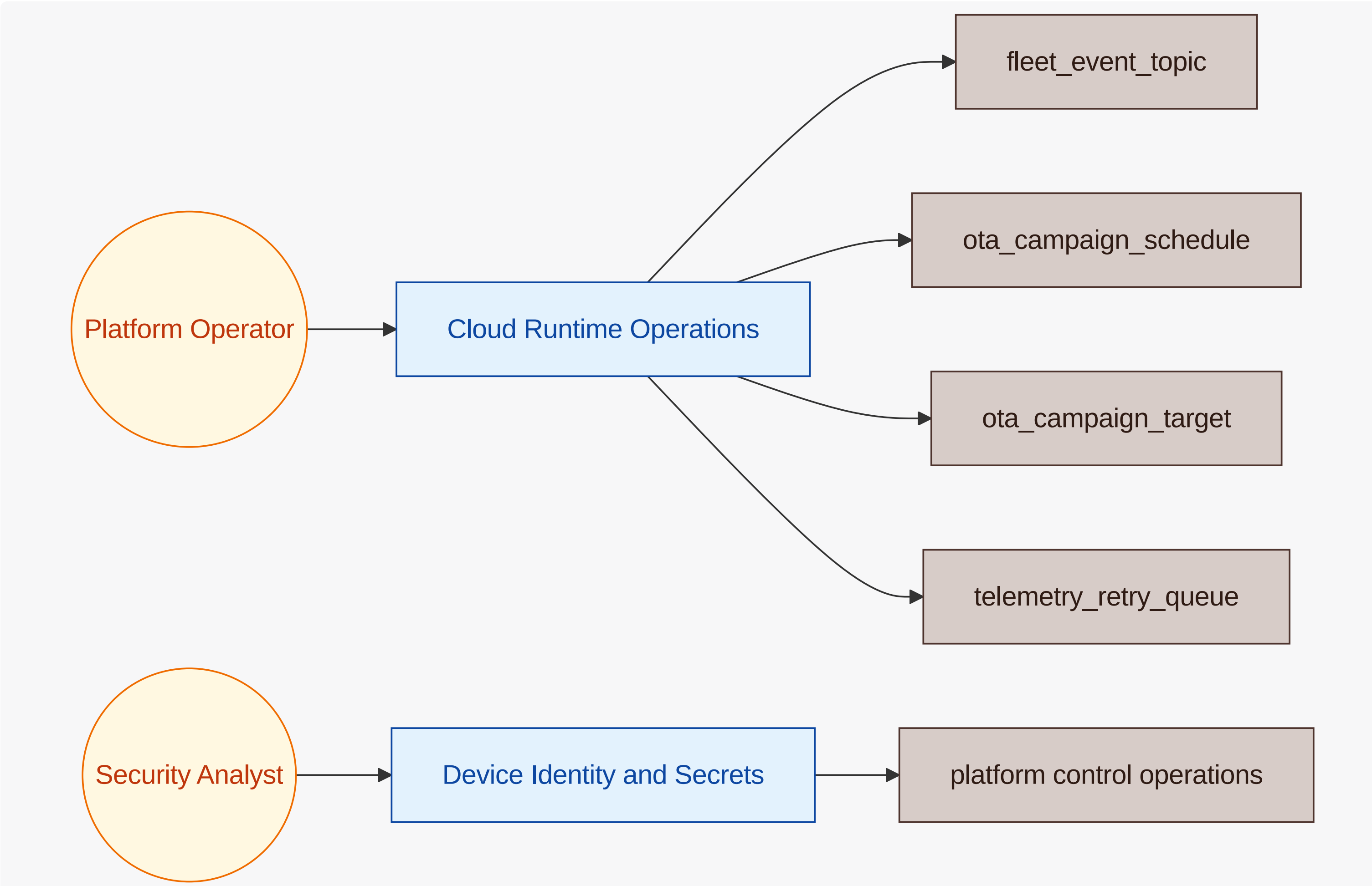
> **NOTE** This graph is inferred from available deployment artifacts (for example Terraform, Flux, Helm) and may be partial when source metadata is incomplete. Namespace and cluster boundaries are rendered as containers when target/placement relationships are inferred. Node labels include role hints (for example `Source`, `Kustomization`, `Release`, `Workload`, `ns/…`) to distinguish similarly named elements.



8.5. Platform Operations Graph

What this diagram shows: which platform actors operate which platform units and the control-plane targets they influence.

> **NOTE** Platform operations links are inferred from authored actor mappings plus discovered deployment control-plane artifacts.



8.6. Functional Groups and Units (View Scoped)

8.6.1. Machine Edge (FG-MACHINE-EDGE)

Machine-edge software artifacts are rolled independently from `cloud control` to isolate device `risk`.

Machine Telemetry Collection (FU-MACHINE-TELEMETRY-COLLECTION)

Machine Telemetry Collection Deployment

Deployed as edge software component with independent release cadence.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-COF-001`, `REQ-COF-006`, `REQ-COF-011`
Deployment dependencies	`Device Identity and Secrets`, `Fleet Ingestion API`, `IoT Ingest Endpoint`, `MQTT Device SDK`
Inferred deployment evidence	code modules: `machine_telemetry`

OTA Update Agent (FU-OTA-UPDATE-AGENT)

OTA Update Agent Deployment

Versioned and rolled independently from `telemetry` pipeline components.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	REQ-COF-003, REQ-COF-004, REQ-COF-005, REQ-COF-010
Deployment dependencies	Device Identity and Secrets, Firmware Bundle Store, Update Campaign Orchestration
Inferred deployment evidence	code modules: ota_update_agent

8.6.2. Cloud Control (FG-CLOUD-CONTROL)

Cloud-control units deploy independently to reduce blast radius during campaign logic changes.

Fleet Ingestion API (FU-FLEET-INGESTION-API)

Fleet Ingestion API Deployment

Deployed as independent cloud ingestion boundary to isolate scaling concerns.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	REQ-COF-001, REQ-COF-002, REQ-COF-006, REQ-COF-008, REQ-COF-009, REQ-COF-011, REQ-COF-012
Deployment dependencies	Cloud Runtime SDK, Device Identity and Secrets, Fleet Observability Reporting
Inferred deployment evidence	runtime: fleet_ingestion_api

Fleet Observability Reporting (FU-FLEET-OBSERVABILITY-REPORTING)

Fleet Observability Reporting Deployment

Deployed as reporting plane with independent scaling and retention controls.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	REQ-COF-002, REQ-COF-005, REQ-COF-007, REQ-COF-008, REQ-COF-009, REQ-COF-010, REQ-COF-012
Deployment dependencies	Cloud Logging Service, Device Identity and Secrets
Inferred deployment evidence	runtime: fleet_observability_reporting

Update Campaign Orchestration (FU-UPDATE-CAMPAIGN-ORCHESTRATION)

Update Campaign Orchestration Deployment

Isolated deployment supports safe OTA policy evolution.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	REQ-COF-003, REQ-COF-005, REQ-COF-010
Deployment dependencies	Device Identity and Secrets, Firmware Signer Service, Fleet Observability Reporting
Inferred deployment evidence	runtime: update_campaign_orchestration

8.6.3. Platform Operations (FG-PLATFORM-OPERATIONS)

`Platform operations` governs release sequencing, runtime placement, and operational guardrails.

Cloud Runtime Operations (FU-CLOUD-RUNTIME-OPERATIONS)

Cloud Runtime Operations Deployment

Owns source-to-runtime release controls and rollback guardrails.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-COF-008`, `REQ-COF-012`
Deployment dependencies	`Cloud Logging Service`, `Device Identity and Secrets`
Inferred deployment evidence	runtime: `fleet_event_topic`, `ota_campaign_schedule`, `ota_campaign_target`, `telemetry_retry_queue`

Device Identity and Secrets (FU-DEVICE-IDENTITY-SECRETS)

Device Identity and Secrets Deployment

Managed as shared control-plane boundary with controlled key rotation.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-COF-004`, `REQ-COF-007`
Deployment dependencies	none
Inferred deployment evidence	code modules: `device_identity_secrets`

9. Security View

Show inferred exposure and dependency `risk` points aligned to unit boundaries, focused on attack paths and security `evidence`.

9.1. What This View Answers

- What are the main `trust boundaries` and attack paths?
- Which units are exposed and what security `evidence` exists?
- Where is the threat `model` incomplete?

9.2. Coverage Gaps

Coverage summary: 6/7 units have direct `evidence` coverage in this `view`.

- `Cloud Runtime Operations` has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.

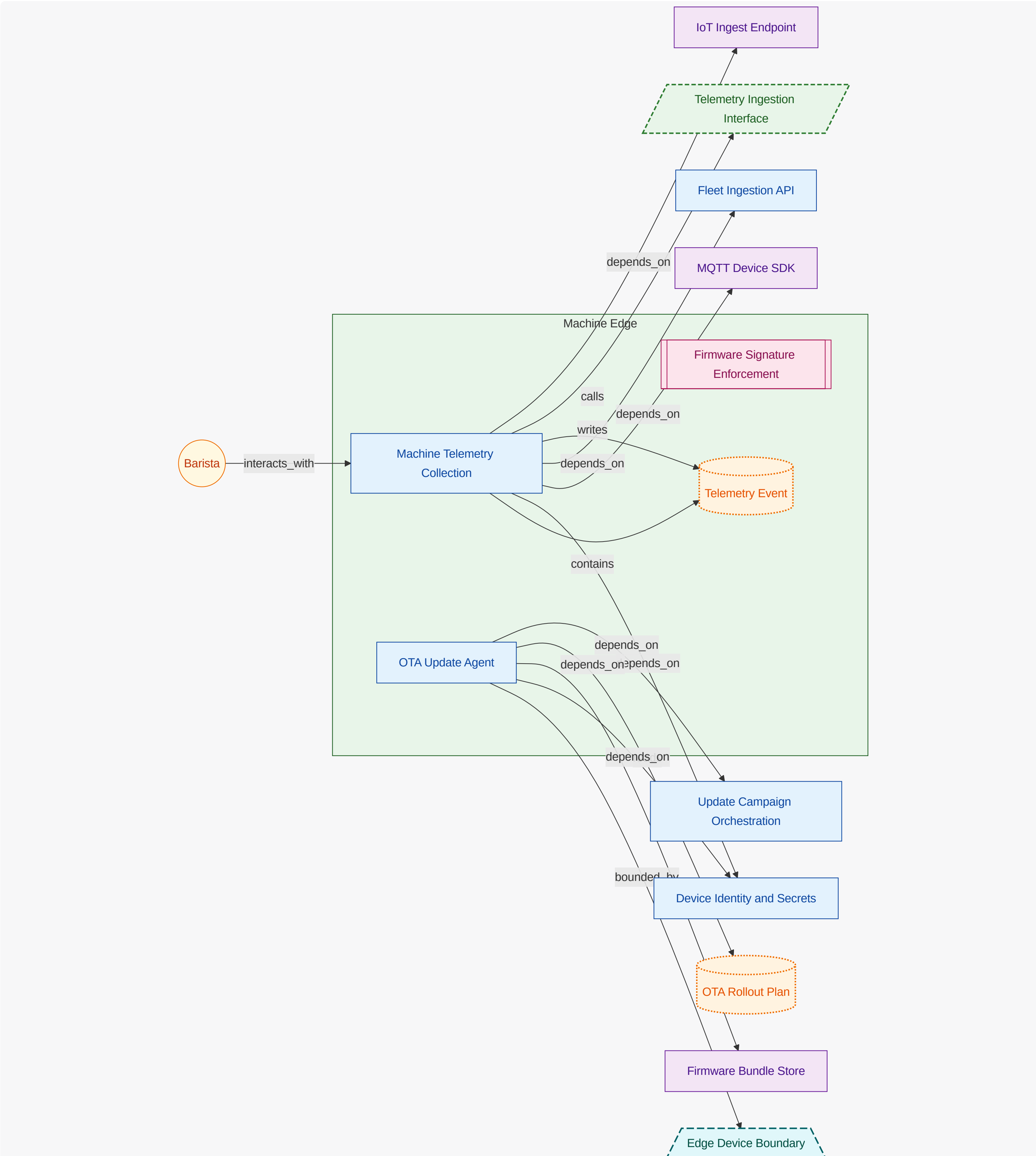
9.3. Recommended Next Evidence Additions

- Add explicit authored attack-vector mappings for units currently marked without threat linkage.
- Add security signal ownership metadata (logs/alerts/`events`) for unresolved observability `evidence`.
- Map additional security-relevant dependencies and exposures into `authored mappings` for traceable coverage.

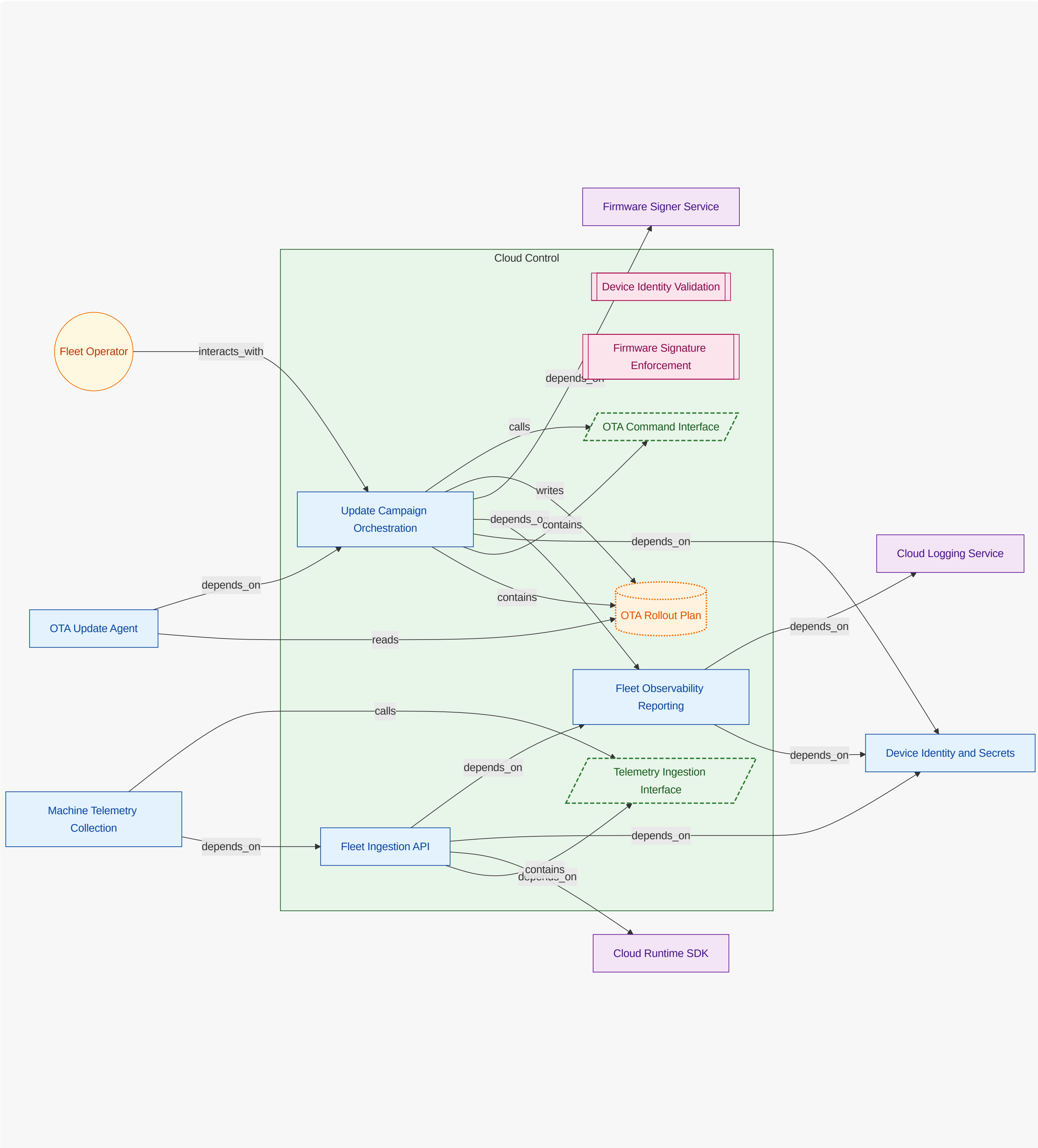
9.4. Threat Model Context Diagram

What these diagrams show: OWASP-style context views split by functional group. Each group diagram keeps owned functional units and owned security-relevant entities inside the functional-group container; connected actors, referenced elements, external functional units, and other external entities are rendered outside it.

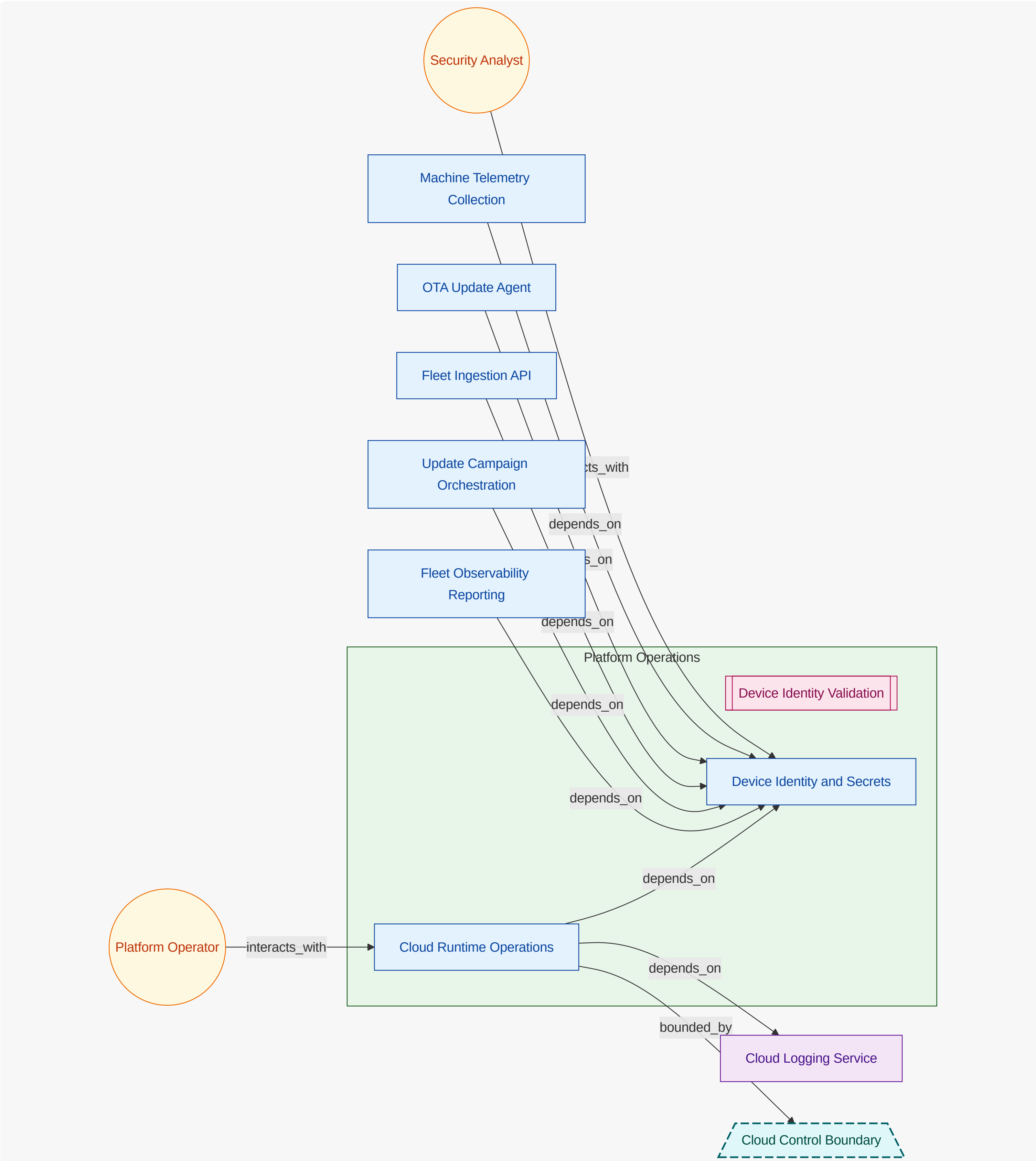
9.4.1. Machine Edge (FG-MACHINE-EDGE)



9.4.2. Cloud Control (FG-CLOUD-CONTROL)



9.4.3. Platform Operations (FG-PLATFORM-OPERATIONS)



9.5. Threat Scenario Register

9.5.1. Audit log tampering suppresses OTA incident visibility (TS-COFFEE-LOG-TAMPERING)

Adversary attempts to alter or suppress security-relevant logs to hide malicious behavior.

Field	Value
Status	identified
Owner	Security Analyst
Attack Vector	Log Tampering
Scope	Cloud Logging Service , Fleet Observability Reporting
Flows	OTA Rollout Interaction Flow , Telemetry Ingestion Interaction Flow
Likelihood	low
Impact	high
Severity	medium
Risk	RISK-COFFEE-LOG-TAMPERING
Controls	Device Identity Validation
Mitigations	MIT-COFFEE-AUDIT-IMMUTABILITY
Verifications	CV-COFFEE-AUDIT-INTEGRITY-ANALYSIS
Operational Notes	Track closure through mitigation and verification linkage.

Control Verification Status

CV-COFFEE-AUDIT-INTEGRITY-ANALYSIS

Field	Value
Control	Device Identity Validation
Method	inspection
Status	partial
Owner	Security Analyst
Last Tested	2026-04-20
Risks	RISK-COFFEE-LOG-TAMPERING
Findings	log write permissions are constrained but immutable retention still being expanded

9.5.2. Telemetry replay inflates ingest and distorts reporting (TS-COFFEE-TELEMETRY-REPLAY)

Replayed telemetry bursts attempt to pollute fleet metrics and trigger false alerts.

Field	Value
Status	mitigating

Field	Value
Owner	Security Analyst
Attack Vector	Telemetry Replay Abuse
Scope	Fleet Ingestion API , Fleet Observability Reporting , Telemetry Event
Flows	Telemetry Ingestion Interaction Flow
Likelihood	medium
Impact	medium
Severity	medium
Risk	RISK-COFFEE-TELEMETRY-REPLAY
Controls	Device Identity Validation
Mitigations	MIT-COFFEE-TELEMETRY-REPLAY-WINDOW
Verifications	CV-COFFEE-TELEMETRY-REPLAY
Operational Notes	Track closure through mitigation and verification linkage.

Control Verification Status

CV-COFFEE-TELEMETRY-REPLAY

Field	Value
Control	Device Identity Validation
Method	analysis
Status	partial
Owner	Security Analyst
Last Tested	2026-04-18
Risks	RISK-COFFEE-TELEMETRY-REPLAY
Findings	replay detection works for standard windows; high-burst tuning still in progress

9.5.3. Tampered firmware bundle executes on edge fleet (TS-COFFEE-MALICIOUS-OTA-BUNDLE)

Attacker attempts to deliver malicious firmware by replacing or tampering rollout artifact.

Field	Value
Status	mitigating
Owner	Security Analyst

Field	Value
Attack Vector	Malicious Firmware Bundle
Scope	Firmware Bundle Store , OTA Update Agent , Update Campaign Orchestration
Flows	OTA Rollout Interaction Flow
Likelihood	medium
Impact	high
Severity	high
Risk	RISK-COFFEE-MALICIOUS-OTA
Controls	Firmware Signature Enforcement
Mitigations	MIT-COFFEE-FW-SIGNATURE-VERIFY
Verifications	CV-COFFEE-FW-SIGNATURE-TEST
Operational Notes	Track closure through mitigation and verification linkage.

Control Verification Status

CV-COFFEE-FW-SIGNATURE-TEST

Field	Value
Control	Firmware Signature Enforcement
Method	test
Status	pass
Owner	Security Analyst
Last Tested	2026-04-12
Risks	RISK-COFFEE-MALICIOUS-OTA
Findings	unsigned or tampered bundles rejected during OTA execution

9.5.4. Spoofed OTA command triggers unauthorized rollout (TS-COFFEE-OTA-COMMAND-SPOOF)

Unauthorized sender attempts to push OTA command messages to device fleet topic.

Field	Value
Status	mitigating
Owner	Security Analyst
Attack Vector	Spoofed Device Identity

Field	Value
Scope	OTA Command Interface, OTA Update Agent, Update Campaign Orchestration
Flows	OTA Rollout Interaction Flow
Likelihood	medium
Impact	high
Severity	high
Risk	RISK-COFFEE-DEVICE-SPOOFING
Controls	Device Identity Validation
Mitigations	MIT-COFFEE-DEVICE-IDENTITY-ENFORCEMENT
Verifications	CV-COFFEE-DEVICE-IDENTITY-TEST
Operational Notes	Track closure through mitigation and verification linkage.

Control Verification Status

CV-COFFEE-DEVICE-IDENTITY-TEST

Field	Value
Control	Device Identity Validation
Method	test
Status	pass
Owner	Security Analyst
Last Tested	2026-04-13
Risks	RISK-COFFEE-DEVICE-SPOOFING
Findings	unauthorized command publishers rejected by identity policy

9.6. Threat Mitigation Coverage

ID	Threat Scenario	Control	Status	Effectiveness	Owner	Verification Links	Notes
MIT-COFFEE-AUDIT-IMMUTABILITY	TS-COFFEE-LOG-TAMPERING	Device Identity Validation	planned	medium	Security Analyst	CV-COFFEE-AUDIT-INTEGRITY-ANALYSIS	Planned immutable log retention and append-only controls for OTA incident trails.

ID	Threat Scenario	Control	Status	Effectiveness	Owner	Verification Links	Notes
MIT-COFFEE-DEVICE-IDENTITY-ENFORCEMENT	TS-COFFEE-OTA-COMMAND-SPOOF	Device Identity Validation	implemented	high	Security Analyst	CV-COFFEE-DEVICE-IDENTITY-TEST	Device identity claims gate command and telemetry channels.
MIT-COFFEE-FW-SIGNATURE-VERIFY	TS-COFFEE-MALICIOUS-OTA-BUNDLE	Firmware Signature Enforcement	verified	high	Security Analyst	CV-COFFEE-FW-SIGNATURE-TEST	OTA agent enforces signature verification with signer metadata checks.
MIT-COFFEE-TELEMETRY-REPLAY-WINDOW	TS-COFFEE-TELEMETRY-REPLAY	Device Identity Validation	partial	medium	Security Analyst	CV-COFFEE-TELEMETRY-REPLAY	Replay window and dedup controls are active with ongoing burst tuning.

9.7. Security Flow Semantics

ID	Title	Kind	Direction	Frequency	Source	Destination	Protocol
FLOW-COFFEE-OTA-ROLLOUT	OTA Rollout Interaction Flow	security	outbound	scheduled	Fleet Operator	OTA Update Agent	mqtt
FLOW-COFFEE-TELEMETRY-INGEST	Telemetry Ingestion Interaction Flow	data	inbound	realtime	Machine Telemetry Collection	Fleet Ingestion API	mqtt

ID	Auth	Encryption	Integrity	Boundary Crossing	Trust Boundary	Threats	Data
FLOW-COFFEE-OTA-ROLLOUT	device-mtls	tls1.3	firmware-signature	true	Edge Device Boundary	TS-COFFEE-MALICIOUS-OTA-BUNDLE, TS-COFFEE-OTA-COMMAND-SPOOF	OTA Rollout Plan
FLOW-COFFEE-TELEMETRY-INGEST	device-identity-token	tls1.3	signed-telemetry-envelope	true	Edge Device Boundary	TS-COFFEE-LOG-TAMPERING, TS-COFFEE-TELEMETRY-REPLAY	Telemetry Event

9.8. Threat Assumptions

ID	Title	Status	Owner	Applies To	Rationale
ASM-COFFEE-DEVICE-ROOT-OF-TRUST	Device trust anchors are provisioned securely	accepted	Security Analyst	Device Identity and Secrets, Edge Device Boundary, OTA Update Agent	Secure provisioning path is audited by platform security and manufacturing controls.

9.9. Threat Model Out-of-Scope Decisions

ID	Title	Status	Owner	Applies To	Expires On	Reason
OOS-COFFEE-HARDWARE-PHYSICAL-TAMPER	Physical tamper resistance of device enclosure	approved	Security Analyst	Coffee Edge Fleet	2027-12-31	Hardware anti-tamper sensor design and physical sealing are owned by hardware product teams.

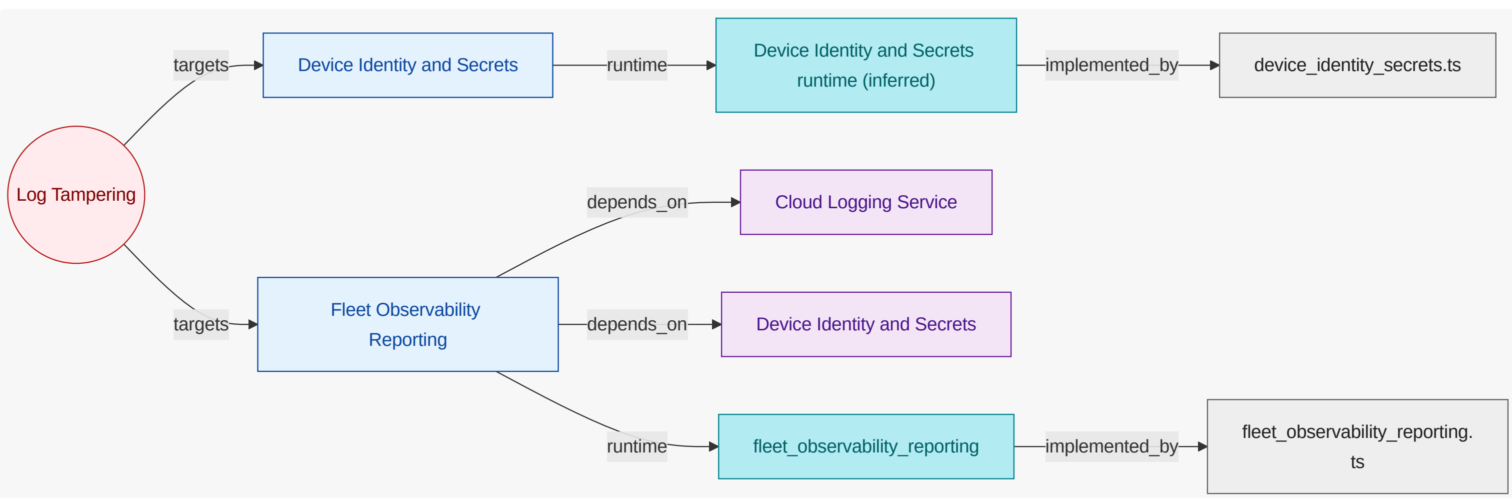
9.10. Attack Vector Chapters

9.10.1. Log Tampering (AV-LOG-TAMPERING)

Attempts to suppress or alter audit/operational logs.

Mitigated by: none

Trust boundaries: none



Device Identity and Secrets (FU-DEVICE-IDENTITY-SECRETS)

Protects credential lifecycle and enforces least-privilege access patterns.

Attack Surface

none

Security Evidence Present

code modules: device_identity_secrets

Fleet Observability Reporting (FU-FLEET-OBSERVABILITY-REPORTING)

Preserves audit integrity and protects sensitive identifiers in persisted records.

Attack Surface

Cloud Logging Service , Device Identity and Secrets

Security Evidence Present

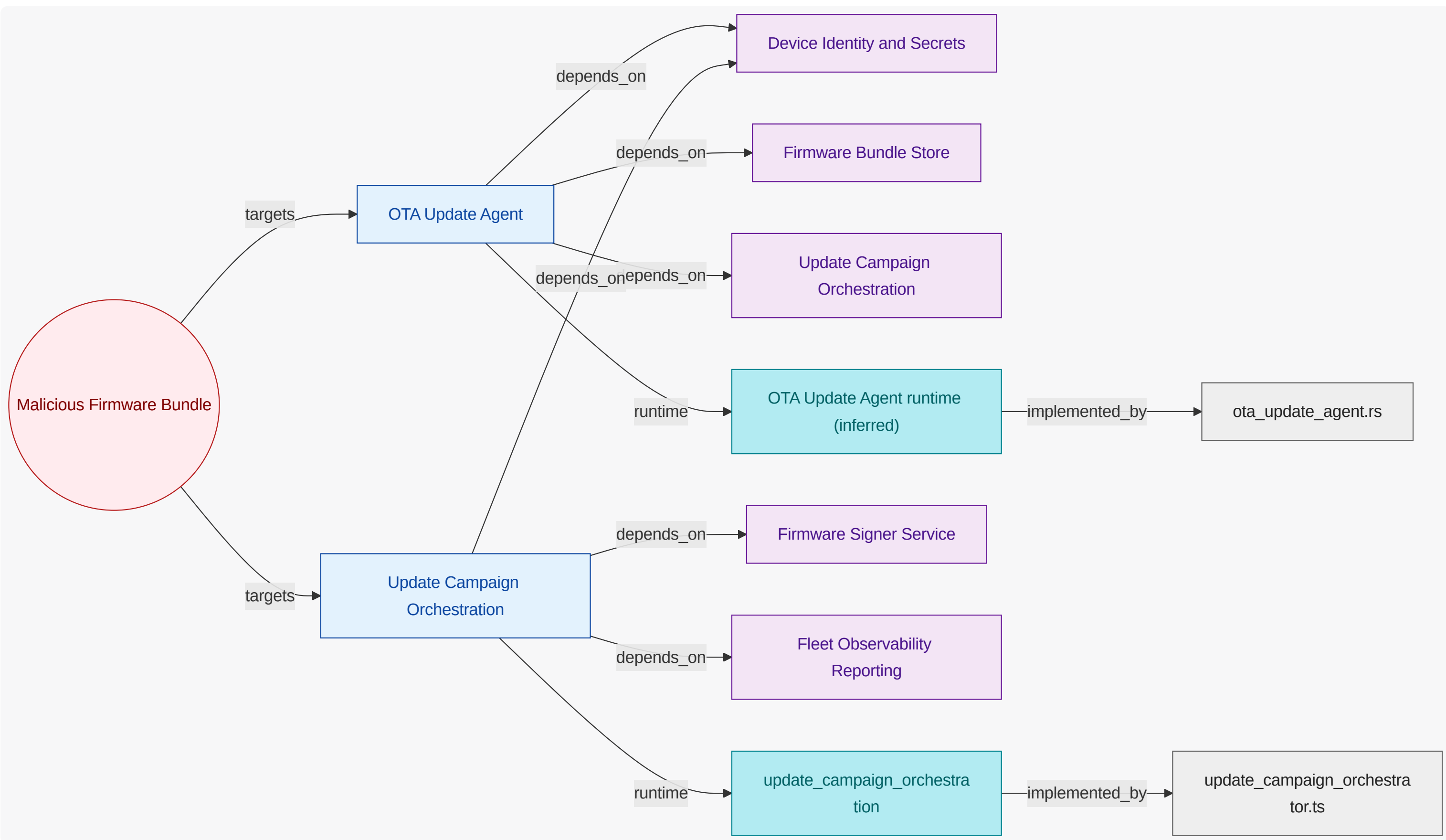
runtime: fleet_observability_reporting | code modules: fleet_observability_reporting

9.10.2. Malicious Firmware Bundle (AV-MALICIOUS-FIRMWARE-BUNDLE)

Tampered OTA payload targeting update agent execution.

Mitigated by: Firmware Signature Enforcement

Trust boundaries: Edge Device Boundary



OTA Update Agent (FU-OTA-UPDATE-AGENT)

Enforces signature validation and rejects untrusted firmware bundles .

Attack Surface

Device Identity and Secrets , Firmware Bundle Store , Update Campaign Orchestration

Security Evidence Present

code modules: ota_update_agent

Update Campaign Orchestration (FU-UPDATE-CAMPAIGN-ORCHESTRATION)

Enforces signed-firmware policy and campaign authorization boundaries.

Attack Surface

Device Identity and Secrets, Firmware Signer Service, Fleet Observability Reporting

Security Evidence Present

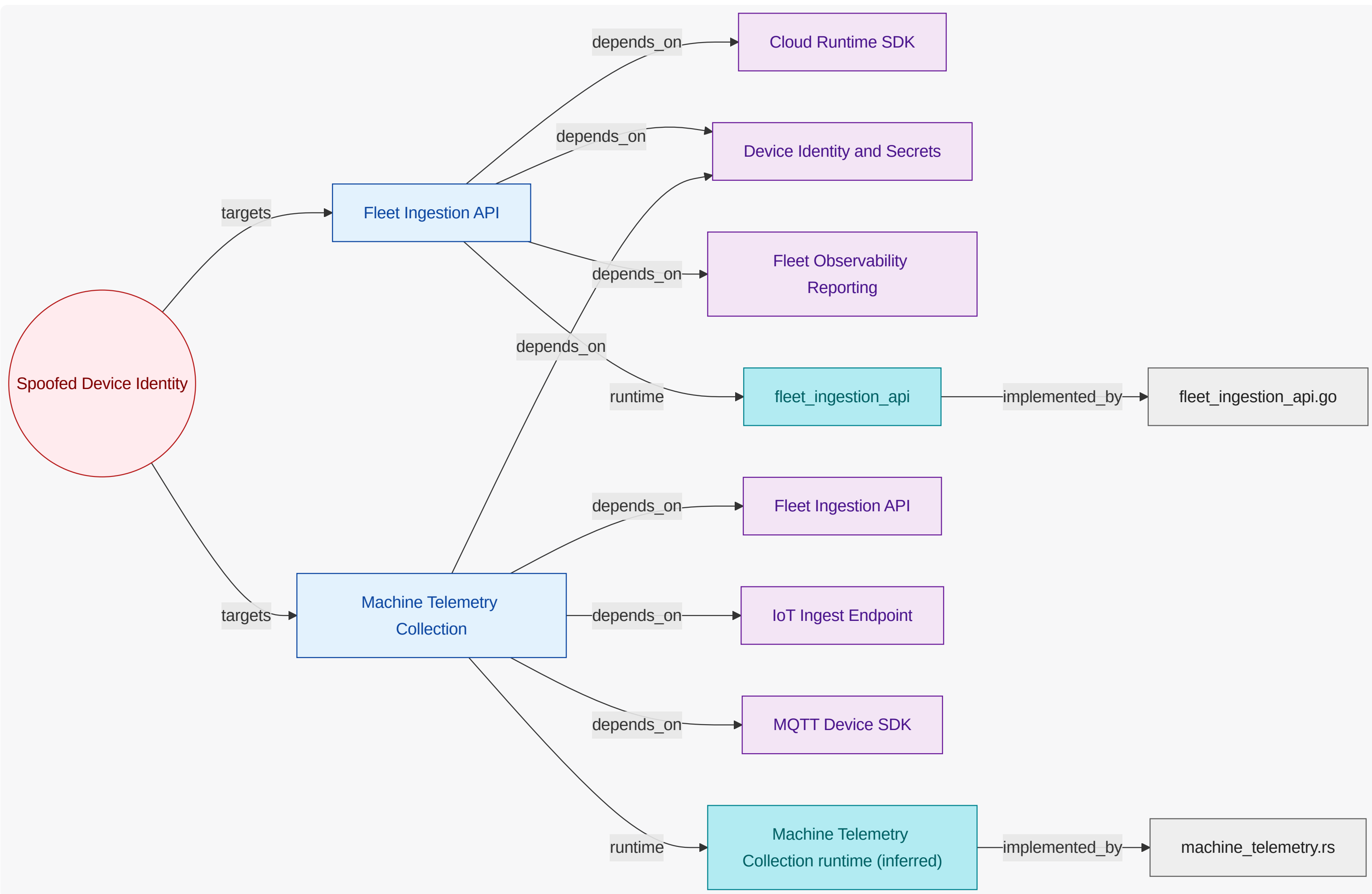
runtime: update_campaign_orchestration | code modules: update_campaign_orchestrator

9.10.3. Spoofed Device Identity (AV-SPOOFED-DEVICE-IDENTITY)

Unauthorized identity attempting telemetry/command access.

Mitigated by: Device Identity Validation

Trust boundaries: none



Fleet Ingestion API (FU-FLEET-INGESTION-API)

Applies identity checks and replay throttling for ingest requests.

Attack Surface

Cloud Runtime SDK, Device Identity and Secrets, Fleet Observability Reporting

Security Evidence Present

runtime: fleet_ingestion_api | code modules: fleet_ingestion_api

Machine Telemetry Collection (FU-MACHINE-TELEMETRY-COLLECTION)

Protects `telemetry` authenticity and defends against `replay abuse` patterns.

Attack Surface

`Device Identity and Secrets`, `Fleet Ingestion API`, `IoT Ingest Endpoint`, `MQTT Device SDK`

Security Evidence Present

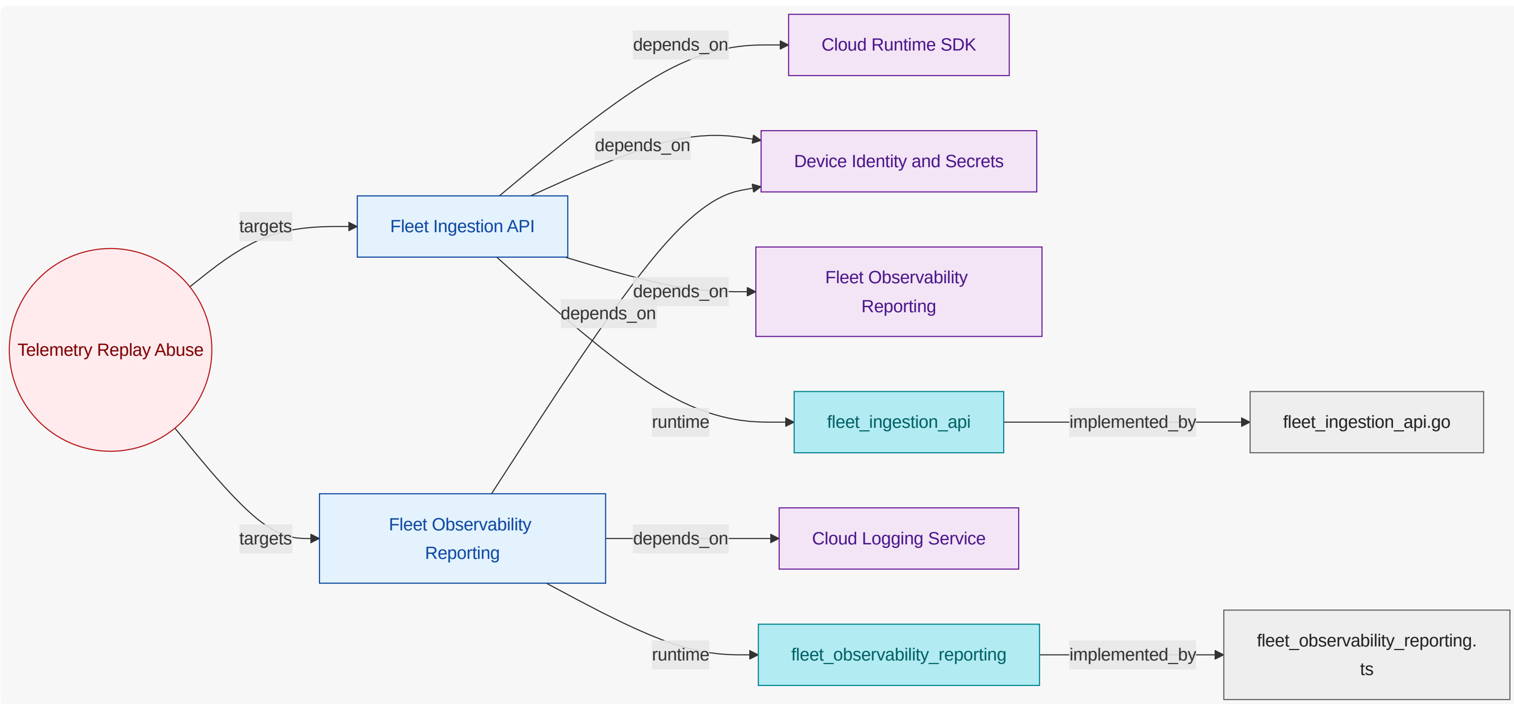
code modules: `machine_telemetry`

9.10.4. Telemetry Replay Abuse (AV-TELEMETRY-REPLAY-ABUSE)

Replayed `telemetry` intended to overload ingestion and corrupt insights.

Mitigated by: none

Trust boundaries: none



Fleet Ingestion API (FU-FLEET-INGESTION-API)

Applies identity checks and replay throttling for ingest requests.

Attack Surface

`Cloud Runtime SDK`, `Device Identity and Secrets`, `Fleet Observability Reporting`

Security Evidence Present

runtime: `fleet_ingestion_api` | code modules: `fleet_ingestion_api`

Fleet Observability Reporting (FU-FLEET-OBSERVABILITY-REPORTING)

Preserves audit integrity and protects sensitive identifiers in persisted records.

Attack Surface

`Cloud Logging Service`, `Device Identity and Secrets`

Security Evidence Present

runtime: `fleet_observability_reporting` | code modules: `fleet_observability_reporting`

10. Traceability View

Show requirement-to-unit-to-evidence traceability, including coverage confidence and explicit `evidence` gaps.

10.1. What This View Answers

- Which `requirements` map to which units?
- Which authored units have runtime/code `evidence` and which do not?
- Where are confidence or completeness gaps in trace coverage?

10.2. Coverage Gaps

Coverage summary: 7/7 units have direct `evidence` coverage in this `view`.

- `Device Identity and Secrets` has no direct inferred runtime `evidence` yet.
- `Machine Telemetry Collection` has no direct inferred runtime `evidence` yet.
- `OTA Update Agent` has no direct inferred runtime `evidence` yet.

10.3. Recommended Next Evidence Additions

- Add explicit requirement-to-owner mappings where `requirement` scope is currently broad or ambiguous.
- Add ownership annotations for runtime and code artifacts where coverage is still unresolved.
- Prioritize closing gaps for units that are authored but lack both runtime and code `evidence`.

11. State Lifecycle View

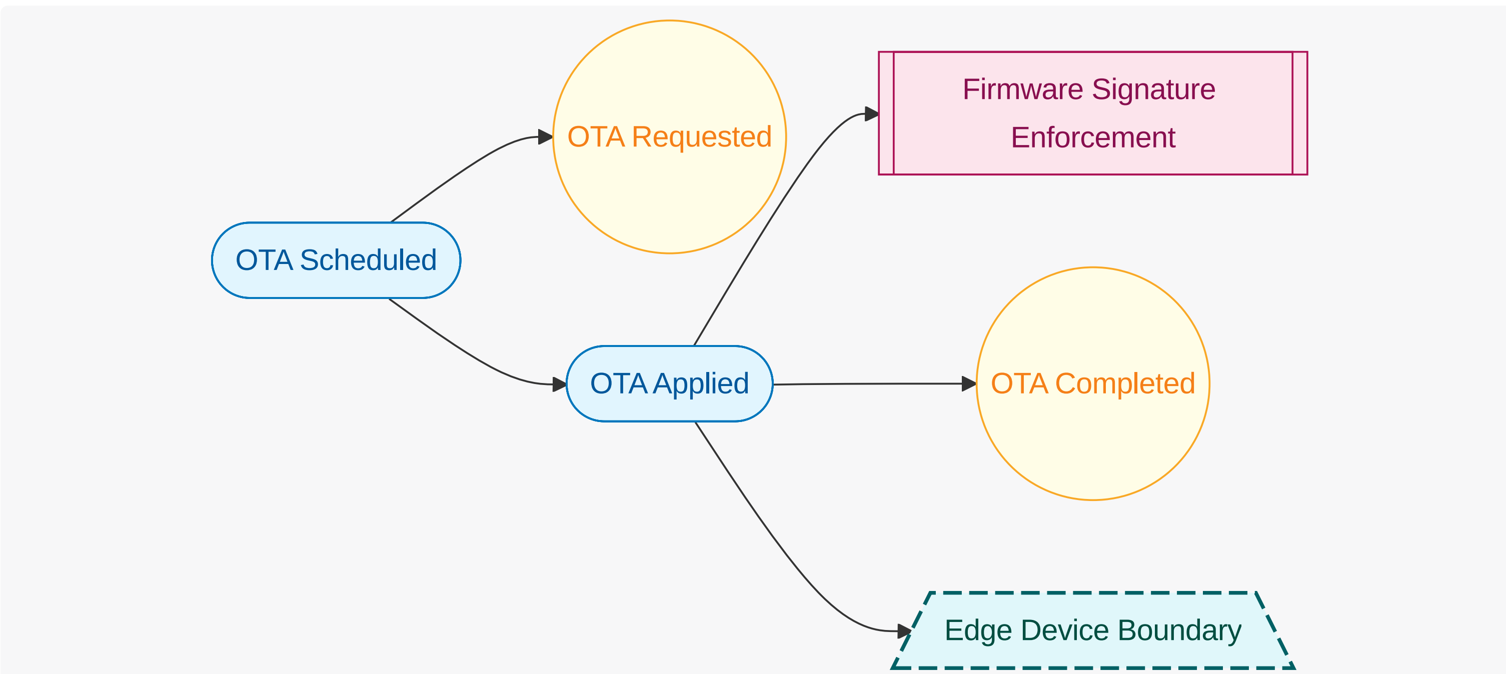
Show workflow `state` transitions, ownership of transitions, and key exceptional paths.

11.1. What This View Answers

- What `states` exist and which `events` trigger transitions?
- Which unit owns each `state` transition?
- Where do automatic and manual flows diverge?

11.2. State Lifecycle Diagram

What this diagram shows: authored lifecycle states, triggering events, guard controls, trust boundaries, and lifecycle transitions for this view.



11.3. Coverage Gaps

Coverage summary: No `functional units` are in scope for this `view`.

- No major coverage gaps detected from current authored and inferred inputs.

11.4. Recommended Next Evidence Additions

- No immediate `evidence` additions are required for this `view`.

11.5. Functional Groups and Units (View Scoped)

12. Interaction Flow View

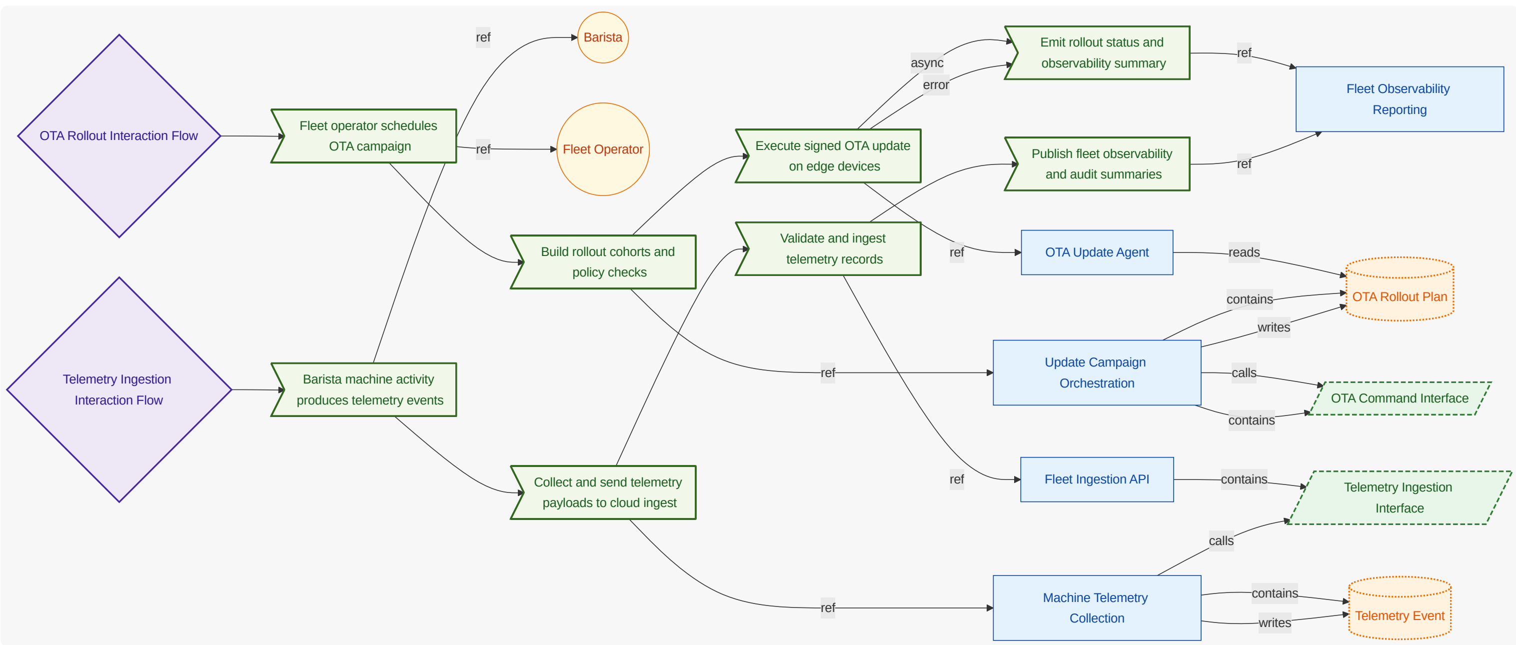
Show user-intent-to-outcome causal `flow`, including ordered steps, data movement, and error/async branches.

12.1. What This View Answers

- What happens from user intent to system outcome, step by step?
- What data enters/exits each step and where are decisions made?
- Where are async and error paths and what completes each path?

12.2. Interaction Flow Diagram

What this diagram shows: authored interaction flow steps, step ordering, async markers, and referenced actors/interfaces/units for this view.



12.3. Coverage Gaps

Coverage summary: 5/5 units have direct `evidence` coverage in this `view`.

- No major coverage gaps detected from current authored and inferred inputs.

12.4. Recommended Next Evidence Additions

- No immediate `evidence` additions are required for this `view`.

12.5. Functional Groups and Units (View Scoped)

12.6. Requirement-to-Unit Mapping (Compact)

What this matrix shows: authored requirement-to-unit mappings in a compact square layout (requirements as rows, functional units as columns).

Requirement	FU-CLOUD-RUNTIME-OPERATIONS	FU-DEVICE-IDENTITY-SECRETS	FU-FLEET-INGESTION-API	FU-FLEET-OBSERVABILITY-REPORTING	FU-MACHINE-TELEMETRY-COLLECTION	FU-OTA-UPDATE-AGENT	FU-UPDATE-CAMPAIGN-ORCHESTRATION
REQ-COF-001			X		X		
REQ-COF-002			X	X			
REQ-COF-003						X	X
REQ-COF-004		X				X	
REQ-COF-005				X		X	X
REQ-COF-006			X		X		
REQ-COF-007		X		X			
REQ-COF-008	X		X	X			
REQ-COF-009			X	X			
REQ-COF-010				X		X	X
REQ-COF-011			X		X		
REQ-COF-012	X		X	X			

12.7. Verification Result Mapping

What this section shows: per-requirement result mapping inferred from test-result artifacts.

Check ID	Requirement	Result	Evidence / Notes
`VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-394794`	`REQ-COF-001`	pass	test-results/contract-telemetry.json; ingest payload accepted with `machine identity` and `telemetry` fields
`VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-B39BBF`	`REQ-COF-002`	not-run	none; No parsed test-result artifact for this `requirement` yet.
`VER-INF-E2E-OTA_ROLLOUT_FLOW-8CBDAC`	`REQ-COF-003`	not-run	none; No parsed test-result artifact for this `requirement` yet.
`VER-INF-UNIT-OTA_CAMPAIGN_TEST-5CC0D8`	`REQ-COF-003`	pass	test-results/unit-ota.xml
`VER-INF-UNIT-OTA_CAMPAIGN_TEST-5CC0D8`	`REQ-COF-004`	pass	test-results/unit-ota.xml
`VER-INF-E2E-OTA_ROLLOUT_FLOW-8CBDAC`	`REQ-COF-005`	not-run	none; No parsed test-result artifact for this `requirement` yet.

Check ID	Requirement	Result	Evidence / Notes
`VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-B39BBF`	`REQ-COF-007`	pass	test-results/integration-logging.json; `audit log record` persisted with redacted `machine identity`

13. Traceability Appendix

13.1. Requirement Details

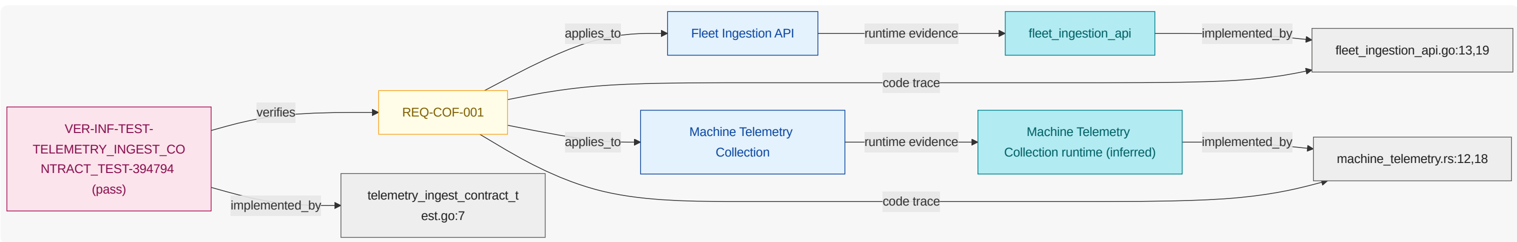
13.1.1. REQ-COF-001

While coffee machine fleet operation is active, when brew cycle completed event is received, the coffee fleet system shall publish brew telemetry to fleet ingestion api for barista operations.

NOTE Primary machine-to-cloud telemetry contract.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



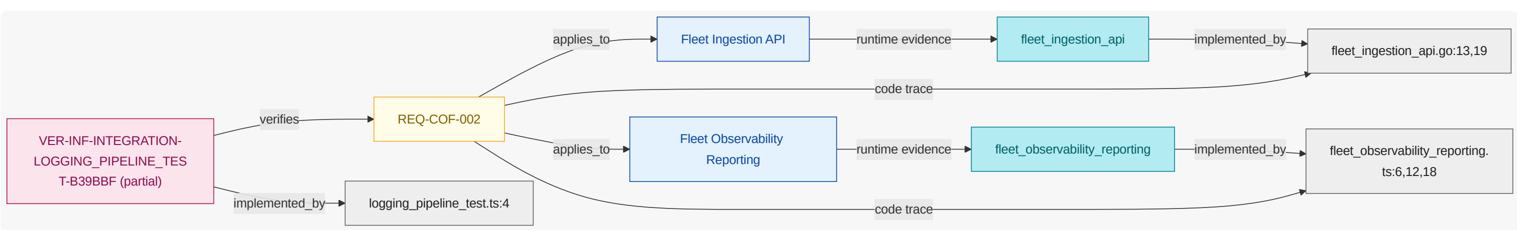
13.1.2. REQ-COF-002

While coffee machine fleet operation is active, when telemetry ingested event is received, the coffee fleet system shall persist fleet telemetry metrics for fleet operator.

NOTE Operational observability requirement.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



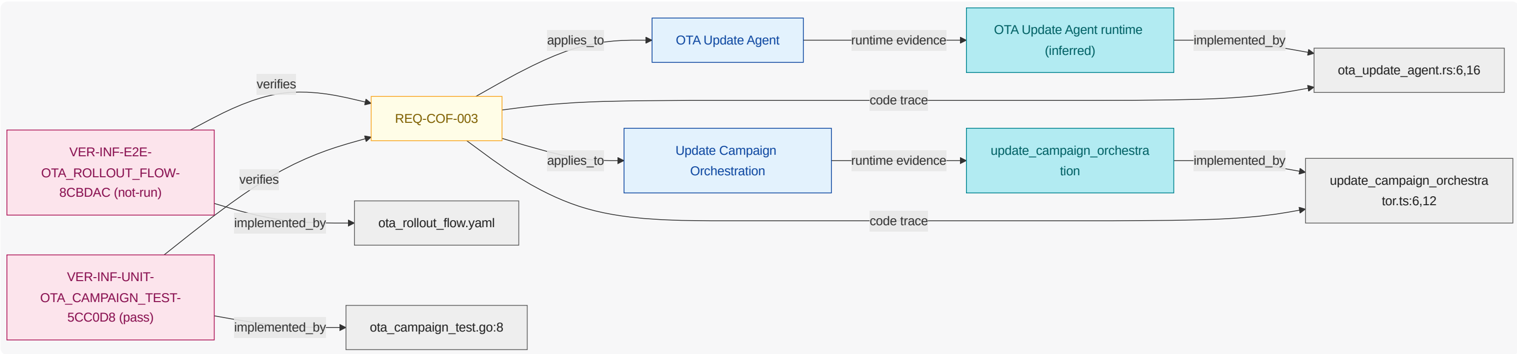
13.1.3. REQ-COF-003

Where ota campaign is enabled, when ota campaign scheduled event is received, the coffee fleet system shall distribute signed firmware bundle to targeted machine cohort using ota campaign policy.

NOTE OTA rollout behavior from central cloud control.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



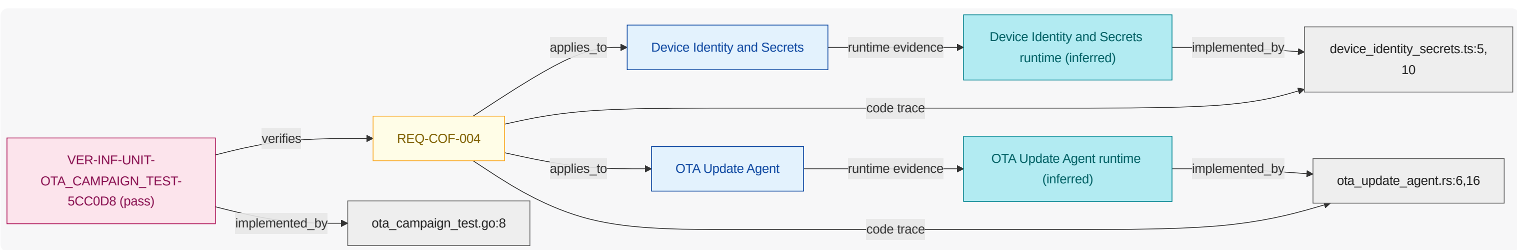
13.1.4. REQ-COF-004

Where `ota campaign is enabled`, when `ota campaign scheduled event is received`, `the coffee fleet system` shall gate `firmware apply` on `firmware signature is valid`.

> **NOTE** Firmware integrity and `trust boundary` `control`.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



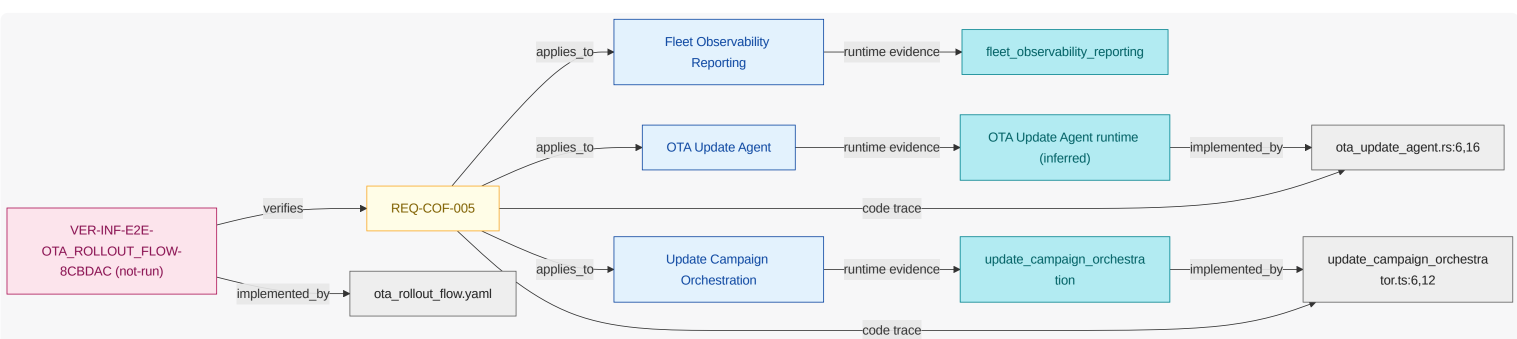
13.1.5. REQ-COF-005

If `firmware validation failed event is received`, then `the coffee fleet system` shall `rollback firmware apply`.

> **NOTE** Rollback and failure signaling path.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



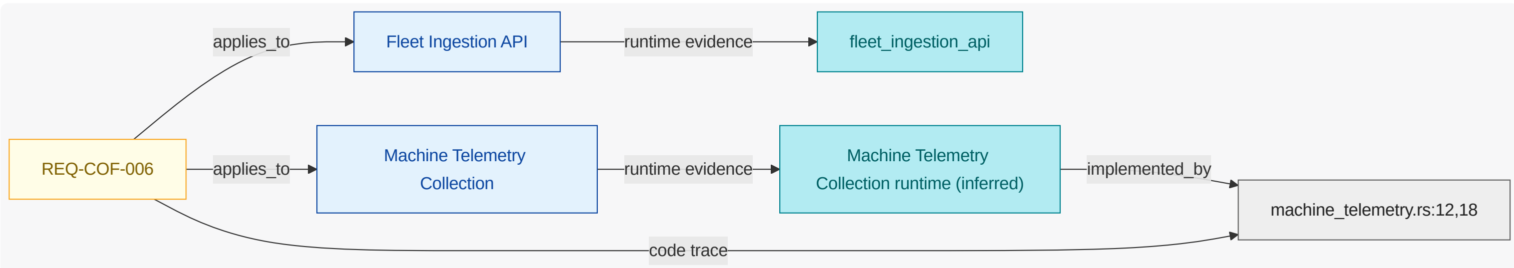
13.1.6. REQ-COF-006

If `cloud ingest endpoint is unavailable`, then `the coffee fleet system` shall queue `telemetry` for retry.

NOTE	Resilience for temporary cloud outage.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



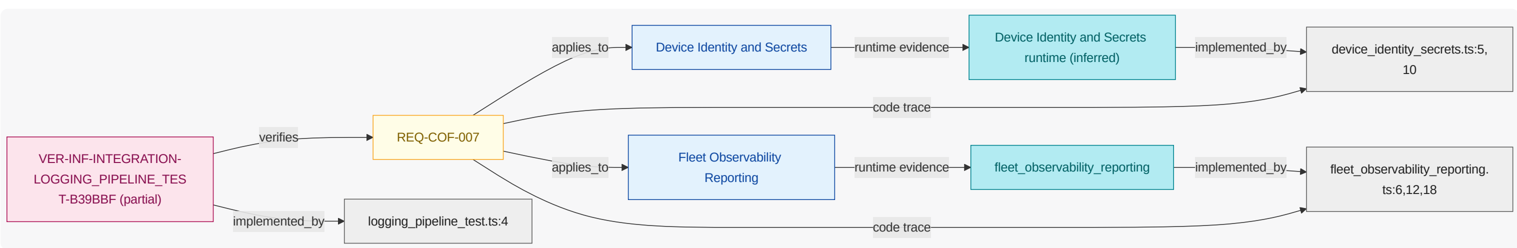
13.1.7. REQ-COF-007

Where `audit logging is enabled`, when `ota status reported event is received`, `the coffee fleet system` shall persist redacted `audit log record` with `machine identity` plus `campaign id`.

NOTE	Audit/compliance logging `requirement`.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



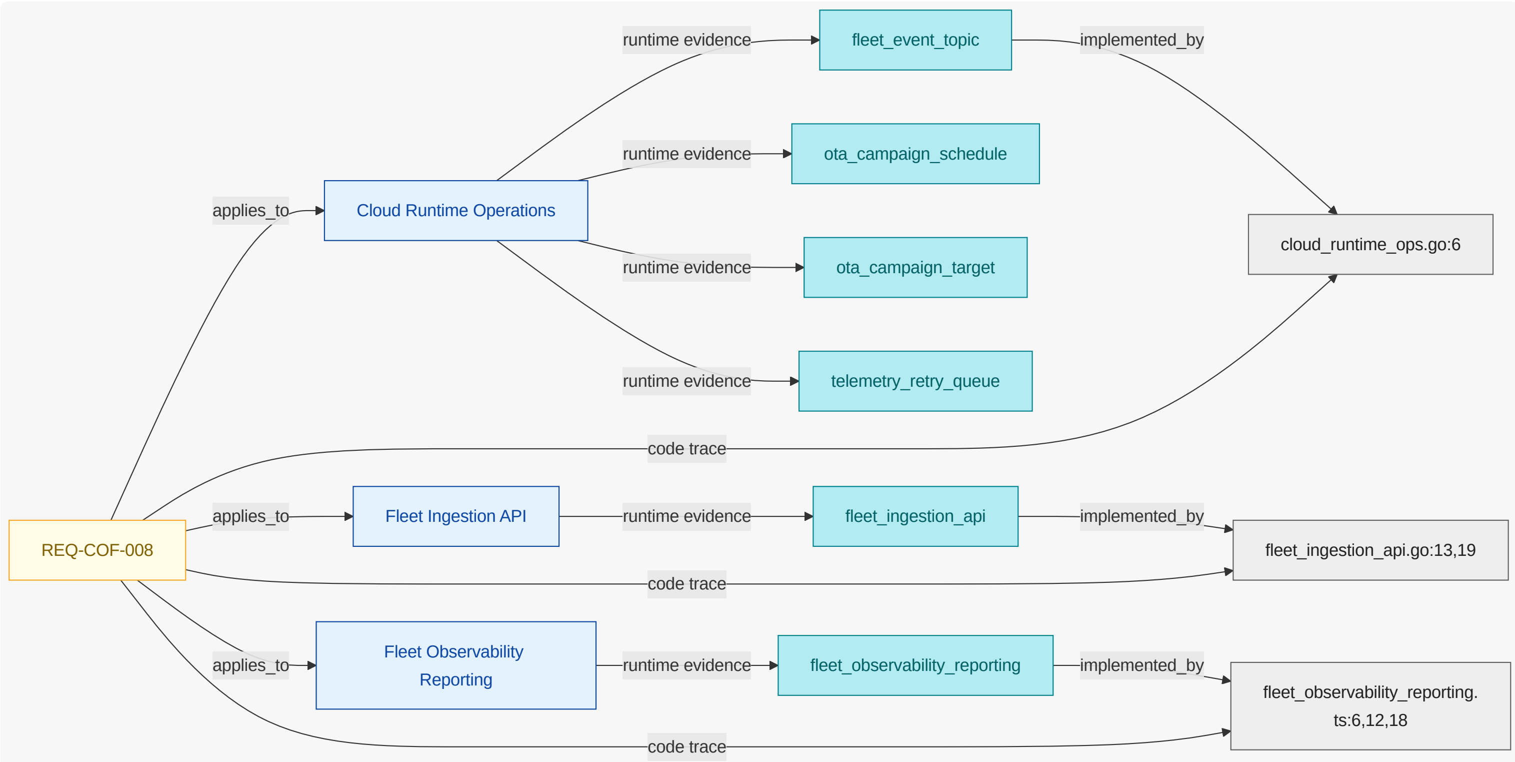
13.1.8. REQ-COF-008

If `telemetry replay abuse is detected`, then `the coffee fleet system` shall throttle duplicate `telemetry`.

NOTE	Abuse `control` and security response `requirement`.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



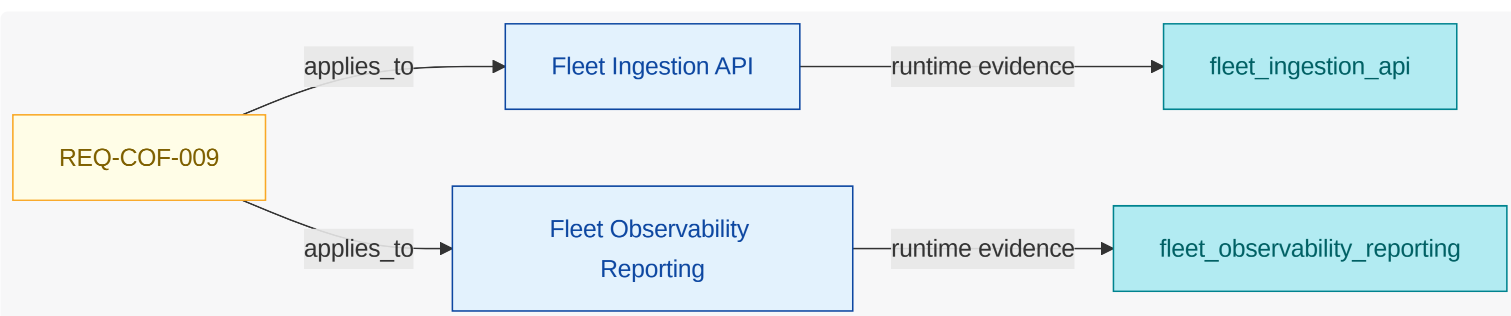
13.1.9. REQ-COF-009

While coffee machine fleet operation is active, when telemetry ingested event is received, the coffee fleet system shall persist logging records for fleet operator.

NOTE Operational logging persistence requirement.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



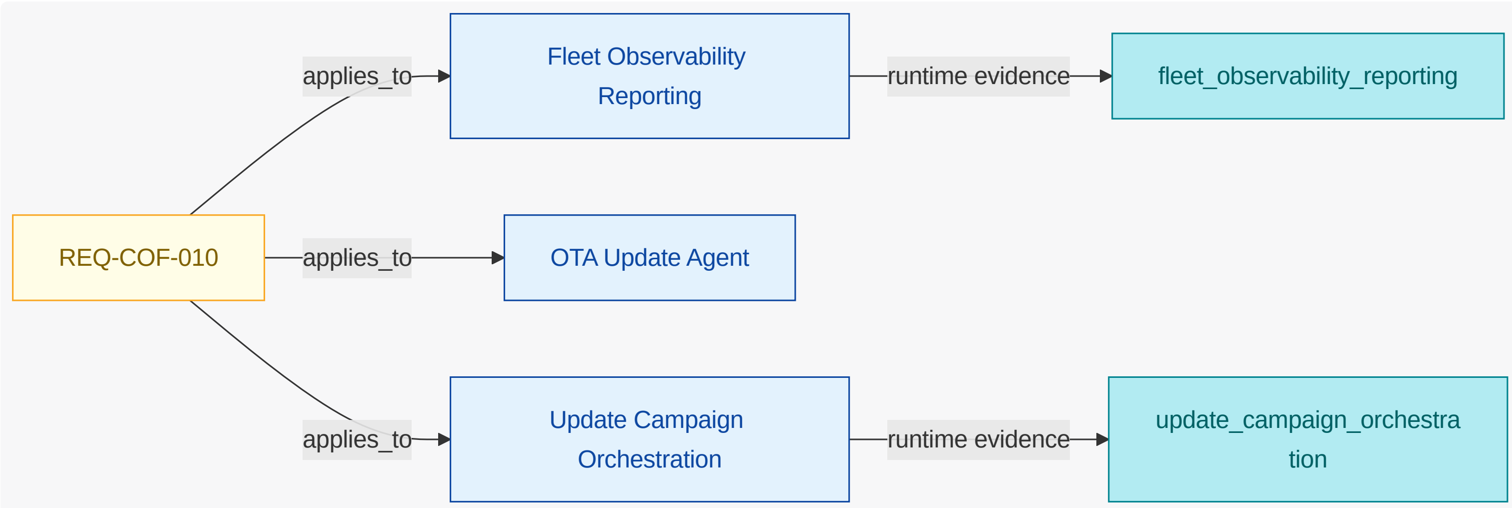
13.1.10. REQ-COF-010

If firmware validation failed event is received, then the coffee fleet system shall report ota status to platform operator.

NOTE OTA failure status signaling requirement.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



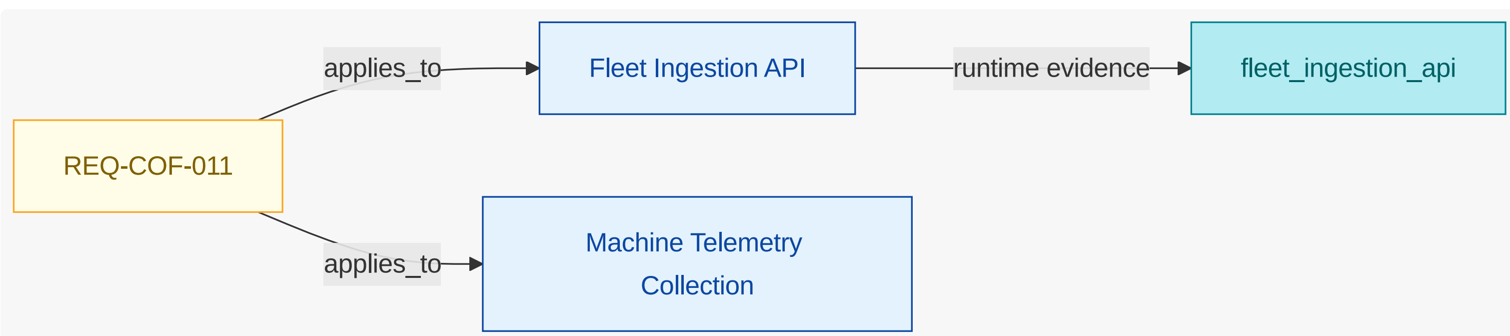
13.1.11. REQ-COF-011

If `cloud ingest endpoint is unavailable`, then `the coffee fleet system` shall flush queued `telemetry` when `normal mode is active`.

> **NOTE** Queued `telemetry` drain behavior after endpoint recovery.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



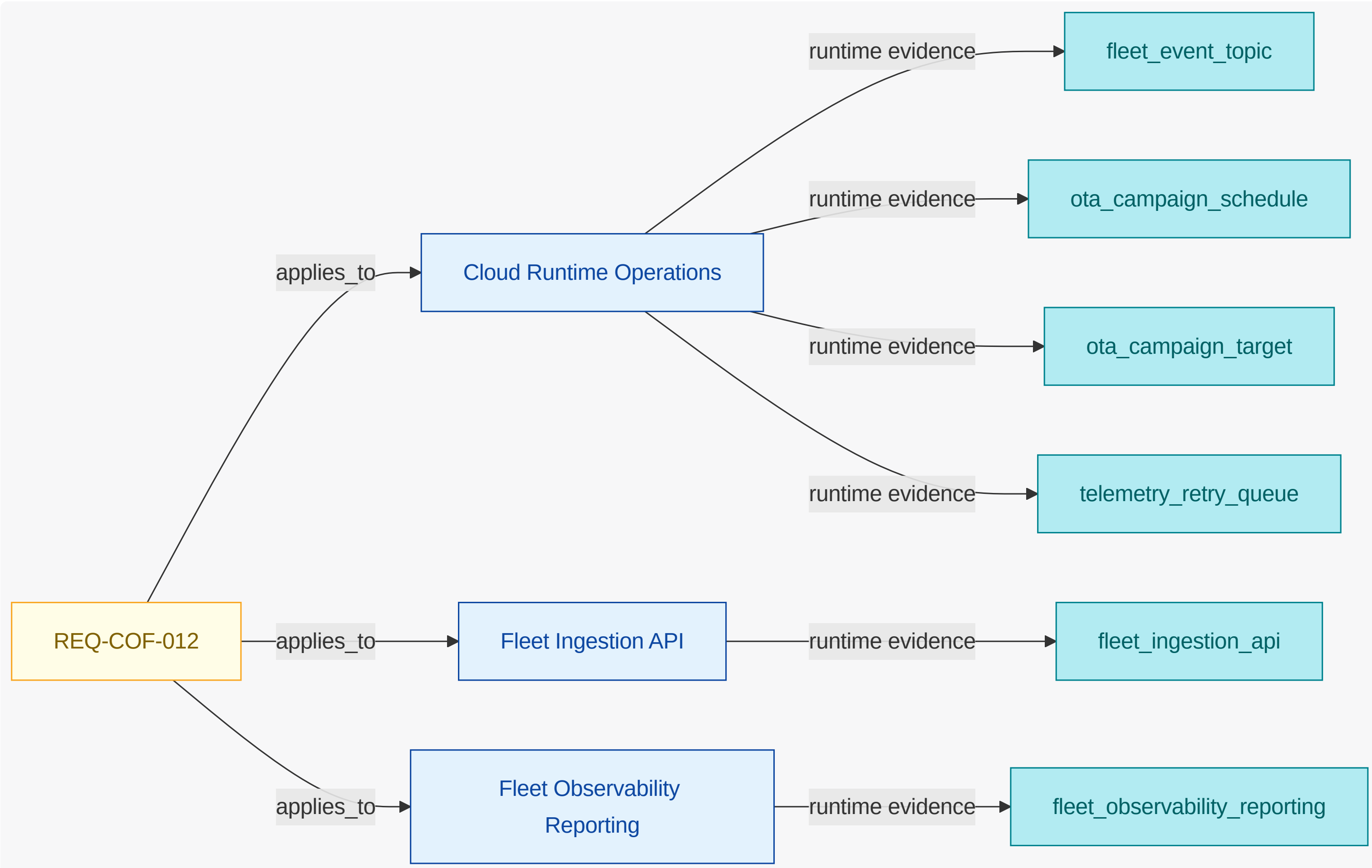
13.1.12. REQ-COF-012

If `telemetry replay abuse is detected`, then `the coffee fleet system` shall notify `security analyst`.

> **NOTE** Security notification `requirement` for `replay abuse` `events`.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



13.2. Verification Inventory

What this section shows: verification checks inferred from test artifacts, linked to test code elements and mapped to verified requirements.

13.2.1. Ota Rollout Flow

Field	Value
ID	`VER-INF-E2E-OTA_ROLLOUT_FLOW-8CBDAC`
Kind	e2e
Status	not-run
Verifies Requirements	`REQ-COF-003`, `REQ-COF-005`
Test Code Elements	tests/e2e/ota_rollout_flow.yaml
Derived Functional Owners	`FU-FLEET-OBSERVABILITY-REPORTING`, `FU-OTA-UPDATE-AGENT`, `FU-UPDATE-CAMPAIGN-ORCHESTRATION`
Result Summary	not-run:2
Evidence	tests/e2e/ota_rollout_flow.yaml
Notes	runs fleet `OTA rollout` scenario including scheduling, deployment, and `telemetry` confirmation

13.2.2. Logging Pipeline Test

Field	Value
ID	`VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-B39BBF`
Kind	integration
Status	partial
Verifies Requirements	`REQ-COF-002`, `REQ-COF-007`
Test Code Elements	logging_pipeline_test.ts:4
Derived Functional Owners	`FU-DEVICE-IDENTITY-SECRETS`, `FU-FLEET-INGESTION-API`, `FU-FLEET-OBSERVABILITY-REPORTING`
Result Summary	not-run:1, pass:1
Evidence	test-results/integration-logging.json, tests/integration/logging_pipeline_test.ts
Notes	validates `telemetry` logging pipeline persistence and operational reporting signals

13.2.3. Telemetry Ingest Contract Test

Field	Value
ID	`VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-394794`
Kind	test
Status	pass
Verifies Requirements	`REQ-COF-001`
Test Code Elements	telemetry_ingest_contract_test.go:7
Derived Functional Owners	`FU-FLEET-INGESTION-API`, `FU-MACHINE-TELEMETRY-COLLECTION`
Result Summary	pass:1
Evidence	test-results/contract-telemetry.json, tests/contract/telemetry_ingest_contract_test.go
Notes	validates `telemetry` ingest contract fields and accepted request semantics

13.2.4. Ota Campaign Test

Field	Value
ID	`VER-INF-UNIT-OTA_CAMPAIGN_TEST-5CC0D8`
Kind	unit
Status	pass

Field	Value
Verifies Requirements	`REQ-COF-003`, `REQ-COF-004`
Test Code Elements	ota_campaign_test.go:8
Derived Functional Owners	`FU-DEVICE-IDENTITY-SECRETS`, `FU-OTA-UPDATE-AGENT`, `FU-UPDATE-CAMPAIGN-ORCHESTRATION`
Result Summary	pass:2
Evidence	test-results/unit-ota.xml, tests/unit/ota_campaign_test.go
Notes	verifies OTA campaign `signature valid`ation and rollout trigger behavior

14. Reference Index

14.1. Authored References

CTRL-COFFEE-DEVICE-IDENTITY

category=identity-access; description=Require authenticated device identity for telemetry and OTA command channels.

Key	Value
Kind	Control
Name	Device Identity Validation
Mentioned In	Security View

CTRL-COFFEE-FIRMWARE-SIGNATURE

category=integrity; description=Require signed firmware bundles before applying OTA updates.

Key	Value
Kind	Control
Name	Firmware Signature Enforcement
Mentioned In	Security View

DO-COFFEE-OTA-PLAN

termRef=DATA-OTA-PLAN; schemaRef=schemas/ota-plan.json; sensitivity=internal

Key	Value
Kind	Data Object
Name	OTA Rollout Plan
Mentioned In	Security View

DO-COFFEE-TELEMETRY-EVENT

termRef=DATA-TELEMETRY-EVENT; schemaRef=schemas/telemetry-event.json; sensitivity=internal

Key	Value
Kind	Data Object
Name	Telemetry Event
Mentioned In	Security View

DEP-COFFEE-CLOUD-PROD

environment=prod; region=us-east-1; account=cloud; cluster=coffee-cloud; namespace=coffee; trustZone=app

Key	Value
Kind	Deployment Target
Name	Coffee Cloud Production
Mentioned In	n/a

DEP-COFFEE-EDGE-FLEET

environment=edge; region=global; account=device; cluster=edge-fleet; namespace=devices; trustZone=edge

Key	Value
Kind	Deployment Target
Name	Coffee Edge Fleet
Mentioned In	Security View

EVT-COFFEE-OTA-COMPLETED

OTA rollout completed for cohort.

Key	Value
Kind	Event
Name	OTA Completed
Mentioned In	n/a

EVT-COFFEE-OTA-REQUESTED

OTA campaign request accepted.

Key	Value
Kind	Event
Name	OTA Requested
Mentioned In	n/a

FLOW-COFFEE-OTA-ROLLOUT

entry=operator-schedules-rollout; exits=report-outcome; steps=4

Key	Value
Kind	Flow
Name	OTA Rollout Interaction Flow
Mentioned In	Security View

FLOW-COFFEE-TELEMETRY-INGEST

entry=capture-machine-telemetry; exits=publish-fleet-observability; steps=4

Key	Value
Kind	Flow
Name	Telemetry Ingestion Interaction Flow
Mentioned In	Security View

FLOW-COFFEE-OTA-ROLLOUT::execute-update

kind=system_action; ref=FU-OTA-UPDATE-AGENT; action=Execute signed OTA update on edge devices; dataIn=ota_rollout_plan; dataOut=; async=true

Key	Value
Kind	Flow Step
Name	Execute signed OTA update on edge devices
Mentioned In	n/a

FLOW-COFFEE-OTA-ROLLOUT::operator-schedules-rollout

kind=user_action; ref=ACT-FLEET-OPERATOR; action=Fleet operator schedules OTA campaign; dataIn=; dataOut=; async=false

Key	Value
Kind	Flow Step
Name	Fleet operator schedules OTA campaign
Mentioned In	n/a

FLOW-COFFEE-OTA-ROLLOUT::plan-campaign

kind=system_action; ref=FU-UPDATE-CAMPAIGN-ORCHESTRATION; action=Build rollout cohorts and policy checks; dataIn=; dataOut=ota_rollout_plan; async=false

Key	Value
Kind	Flow Step
Name	Build rollout cohorts and policy checks
Mentioned In	n/a

FLOW-COFFEE-OTA-ROLLOUT::report-outcome

kind=system_action; ref=FU-FLEET-OBSERVABILITY-REPORTING; action=Emit rollout status and observability summary; dataIn=; dataOut=; async=false

Key	Value
Kind	Flow Step
Name	Emit rollout status and observability summary
Mentioned In	n/a

FLOW-COFFEE-TELEMETRY-INGEST::capture-machine-telemetry

kind=user_action; ref=ACT-BARISTA; action=Barista machine activity produces telemetry events; dataIn=; dataOut=; async=false

Key	Value
Kind	Flow Step
Name	Barista machine activity produces telemetry events
Mentioned In	n/a

FLOW-COFFEE-TELEMETRY-INGEST::collect-and-send-telemetry

kind=system_action; ref=FU-MACHINE-TELEMETRY-COLLECTION; action=Collect and send telemetry payloads to cloud ingest; dataIn=; dataOut=telemetry_event; async=false

Key	Value
Kind	Flow Step
Name	Collect and send telemetry payloads to cloud ingest
Mentioned In	n/a

FLOW-COFFEE-TELEMETRY-INGEST::ingest-telemetry

kind=system_action; ref=FU-FLEET-INGESTION-API; action=Validate and ingest telemetry records; dataIn=telemetry_event; dataOut=; async=false

Key	Value
Kind	Flow Step
Name	Validate and ingest telemetry records
Mentioned In	n/a

FLOW-COFFEE-TELEMETRY-INGEST::publish-fleet-observability

kind=system_action; ref=FU-FLEET-OBSERVABILITY-REPORTING; action=Publish fleet observability and audit summaries; dataIn=telemetry_event; dataOut=; async=false

Key	Value
Kind	Flow Step

Key	Value
Name	Publish fleet observability and audit summaries
Mentioned In	n/a

IF-COFFEE-OTA-COMMAND

protocol=mqtt; endpoint=mqtt://devices/ota/commands; schemaRef=schemas/ota-command.json; owner=FU-UPDATE-CAMPAIGN-ORCHESTRATION

Key	Value
Kind	Interface
Name	OTA Command Interface
Mentioned In	Security View

IF-COFFEE-TELEMETRY-INGEST

protocol=mqtt; endpoint=mqtt://ingest/topics/telemetry; schemaRef=schemas/telemetry.json; owner=FU-FLEET-INGESTION-API

Key	Value
Kind	Interface
Name	Telemetry Ingestion Interface
Mentioned In	n/a

REQ-COF-001

While coffee machine fleet operation is active, when brew cycle completed event is received, the coffee fleet system shall publish brew telemetry to fleet ingestion api for barista operations.

Key	Value
Kind	Requirement
Name	REQ-COF-001
Mentioned In	Architecture Intent View , Deployment View , Interaction Flow View , Telemetry Ingest Contract Test

REQ-COF-002

While coffee machine fleet operation is active, when telemetry ingested event is received, the coffee fleet system shall persist fleet telemetry metrics for fleet operator.

Key	Value
Kind	Requirement
Name	REQ-COF-002

Key	Value
Mentioned In	Architecture Intent View , Deployment View , Interaction Flow View , Logging Pipeline Test

REQ-COF-003

Where ota campaign is enabled, when ota campaign scheduled event is received, the coffee fleet system shall distribute signed firmware bundle to targeted machine cohort using ota campaign policy.

Key	Value
Kind	Requirement
Name	REQ-COF-003
Mentioned In	Architecture Intent View , Deployment View , Interaction Flow View , Ota Rollout Flow , Ota Campaign Test

REQ-COF-004

Where ota campaign is enabled, when ota campaign scheduled event is received, the coffee fleet system shall gate firmware apply on firmware signature is valid.

Key	Value
Kind	Requirement
Name	REQ-COF-004
Mentioned In	Architecture Intent View , Deployment View , Interaction Flow View , Ota Campaign Test

REQ-COF-005

If firmware validation failed event is received, then the coffee fleet system shall rollback firmware apply.

Key	Value
Kind	Requirement
Name	REQ-COF-005
Mentioned In	Architecture Intent View , Deployment View , Interaction Flow View , Ota Rollout Flow

REQ-COF-006

If cloud ingest endpoint is unavailable, then the coffee fleet system shall queue telemetry for retry.

Key	Value
Kind	Requirement
Name	REQ-COF-006
Mentioned In	Architecture Intent View , Deployment View

REQ-COF-007

Where audit logging is enabled, when ota status reported event is received, the coffee fleet system shall persist redacted audit log record with machine identity plus campaign id.

Key	Value
Kind	Requirement
Name	REQ-COF-007
Mentioned In	Architecture Intent View , Deployment View , Interaction Flow View , Logging Pipeline Test

REQ-COF-008

If telemetry replay abuse is detected, then the coffee fleet system shall throttle duplicate telemetry.

Key	Value
Kind	Requirement
Name	REQ-COF-008
Mentioned In	Architecture Intent View , Deployment View

REQ-COF-009

While coffee machine fleet operation is active, when telemetry ingested event is received, the coffee fleet system shall persist logging records for fleet operator.

Key	Value
Kind	Requirement
Name	REQ-COF-009
Mentioned In	Architecture Intent View , Deployment View

REQ-COF-010

If firmware validation failed event is received, then the coffee fleet system shall report ota status to platform operator.

Key	Value
Kind	Requirement
Name	REQ-COF-010
Mentioned In	Architecture Intent View , Deployment View

REQ-COF-011

If cloud ingest endpoint is unavailable, then the coffee fleet system shall flush queued telemetry when normal mode is active.

Key	Value
Kind	Requirement
Name	REQ-COF-011
Mentioned In	Architecture Intent View , Deployment View

REQ-COF-012

If telemetry replay abuse is detected, then the coffee fleet system shall notify security analyst.

Key	Value
Kind	Requirement
Name	REQ-COF-012
Mentioned In	Architecture Intent View , Deployment View

STATE-COFFEE-OTA-APPLIED

Firmware update successfully applied to target cohort.

Key	Value
Kind	State
Name	OTA Applied
Mentioned In	n/a

STATE-COFFEE-OTA-SCHEDULED

Campaign prepared and waiting for rollout execution.

Key	Value
Kind	State
Name	OTA Scheduled
Mentioned In	n/a

TB-COFFEE-CLOUD-CONTROL

Boundary between cloud app workloads and privileged platform controls.

Key	Value
Kind	Trust Boundary
Name	Cloud Control Boundary
Mentioned In	n/a

TB-COFFEE-EDGE-DEVICE

Boundary between managed edge devices and cloud control plane.

Key	Value
Kind	Trust Boundary
Name	Edge Device Boundary
Mentioned In	Security View

14.2. Catalog References

ACT-BARISTA

Operates coffee machines and observes local machine health.

Key	Value
Kind	Catalog Actor
Name	barista
Mentioned In	REQ-COF-001

ACT-FLEET-OPERATOR

Owns OTA rollout campaigns and fleet-level reliability.

Key	Value
Kind	Catalog Actor
Name	fleet operator
Mentioned In	Security View, REQ-COF-002, REQ-COF-009

ACT-PLATFORM-OPERATOR

Owns runtime deployment and operational controls.

Key	Value
Kind	Catalog Actor
Name	platform operator
Mentioned In	REQ-COF-010

ACT-SECURITY-ANALYST

Reviews identity, firmware integrity, and logging controls.

Key	Value
Kind	Catalog Actor
Name	security analyst
Mentioned In	Security View, REQ-COF-012

apply status

Event raised when device reports OTA apply status.

Key	Value
Kind	Catalog Alias
Name	apply status
Mentioned In	n/a

apply/rollback status

Event raised when device reports OTA apply status.

Key	Value
Kind	Catalog Alias
Name	apply/rollback status
Mentioned In	n/a

audit enabled

Feature flag enabling audit-grade log persistence.

Key	Value
Kind	Catalog Alias
Name	audit enabled
Mentioned In	n/a

audit records

Immutable log record for security and compliance analysis.

Key	Value
Kind	Catalog Alias
Name	audit records
Mentioned In	Architecture Intent View

audit tampering

Attempts to suppress or alter security/operational log records.

Key	Value
Kind	Catalog Alias
Name	audit tampering
Mentioned In	n/a

brew completed

Event raised when a coffee machine completes a brew cycle.

Key	Value
Kind	Catalog Alias
Name	brew completed
Mentioned In	n/a

brew cycle event

Event raised when a coffee machine completes a brew cycle.

Key	Value
Kind	Catalog Alias
Name	brew cycle event
Mentioned In	n/a

campaign commands

Event raised when a fleet OTA campaign is scheduled.

Key	Value
Kind	Catalog Alias
Name	campaign commands
Mentioned In	n/a

campaign identifier

Stable OTA campaign identifier used to correlate rollout and audit records.

Key	Value
Kind	Catalog Alias

Key	Value
Name	campaign identifier
Mentioned In	n/a

campaign ids

Stable OTA campaign identifier used to correlate rollout and audit records.

Key	Value
Kind	Catalog Alias
Name	campaign ids
Mentioned In	n/a

campaign orchestration

Selects target machine groups and coordinates OTA rollout phases.

Key	Value
Kind	Catalog Alias
Name	campaign orchestration
Mentioned In	n/a

campaign policy

Policy describing rollout cohorts, windows, and rollback criteria.

Key	Value
Kind	Catalog Alias
Name	campaign policy
Mentioned In	n/a

campaign scheduled

Event raised when a fleet OTA campaign is scheduled.

Key	Value
Kind	Catalog Alias
Name	campaign scheduled
Mentioned In	n/a

central cloud site

Capability area for fleet ingestion, campaign orchestration, and cloud reporting.

Key	Value
Kind	Catalog Alias
Name	central cloud site
Mentioned In	n/a

cloud sdk

SDK used by cloud units for runtime integrations.

Key	Value
Kind	Catalog Alias
Name	cloud sdk
Mentioned In	n/a

coffee machine edge

Capability area for device-local telemetry and OTA execution behavior.

Key	Value
Kind	Catalog Alias
Name	coffee machine edge
Mentioned In	n/a

device identity

Device identity material used for authenticated publish, ingest, and audit attribution.

Key	Value
Kind	Catalog Alias
Name	device identity
Mentioned In	Security View

device secrets

Manages device identity material and secure configuration.

Key	Value
Kind	Catalog Alias
Name	device secrets

Key	Value
Mentioned In	n/a

device spoofing

Unauthorized actor impersonates a machine identity.

Key	Value
Kind	Catalog Alias
Name	device spoofing
Mentioned In	n/a

device update agent

Downloads, validates, and applies signed firmware updates on machines.

Key	Value
Kind	Catalog Alias
Name	device update agent
Mentioned In	n/a

endpoint outage

Event raised when cloud ingest endpoint becomes unavailable for telemetry ingestion.

Key	Value
Kind	Catalog Alias
Name	endpoint outage
Mentioned In	n/a

firmware apply

Device-level status for firmware download, apply, and rollback outcomes.

Key	Value
Kind	Catalog Alias
Name	firmware apply
Mentioned In	`REQ-COF-004`

firmware artifact

Signed OTA firmware artifact distributed to machine cohorts for update apply.

Key	Value
Kind	Catalog Alias
Name	firmware artifact
Mentioned In	n/a

firmware bundle

Signed OTA firmware artifact distributed to machine cohorts for update apply.

Key	Value
Kind	Catalog Alias
Name	firmware bundle
Mentioned In	Security View

firmware bundles

Signed OTA firmware artifact distributed to machine cohorts for update apply.

Key	Value
Kind	Catalog Alias
Name	firmware bundles
Mentioned In	n/a

firmware store

Object store containing signed OTA firmware bundles.

Key	Value
Kind	Catalog Alias
Name	firmware store
Mentioned In	n/a

firmware tampering

Tampered firmware payload attempts to execute on devices.

Key	Value
Kind	Catalog Alias
Name	firmware tampering
Mentioned In	n/a

fleet admin

Owns OTA rollout campaigns and fleet-level reliability.

Key	Value
Kind	Catalog Alias
Name	fleet admin
Mentioned In	n/a

fleet events

Event raised when cloud ingestion accepts telemetry.

Key	Value
Kind	Catalog Alias
Name	fleet events
Mentioned In	n/a

fleet operation active

State where machines and central cloud are actively serving operations.

Key	Value
Kind	Catalog Alias
Name	fleet operation active
Mentioned In	n/a

fleet platform operations

Capability area for runtime operations and device identity/secrets lifecycle.

Key	Value
Kind	Catalog Alias
Name	fleet platform operations
Mentioned In	n/a

ingest endpoint unavailable

Event raised when cloud ingest endpoint becomes unavailable for telemetry ingestion.

Key	Value
Kind	Catalog Alias

Key	Value
Name	ingest endpoint unavailable
Mentioned In	n/a

ingestion api

Ingests telemetry and machine status from edge devices.

Key	Value
Kind	Catalog Alias
Name	ingestion api
Mentioned In	n/a

ingestion events

Event raised when cloud ingestion accepts telemetry.

Key	Value
Kind	Catalog Alias
Name	ingestion events
Mentioned In	Architecture Intent View

iot endpoint

Cloud endpoint for receiving telemetry and status from devices.

Key	Value
Kind	Catalog Alias
Name	iot endpoint
Mentioned In	n/a

logging records

Immutable log record for security and compliance analysis.

Key	Value
Kind	Catalog Alias
Name	logging records
Mentioned In	REQ-COF-009

logging service

Managed log and metrics service used for fleet observability.

Key	Value
Kind	Catalog Alias
Name	logging service
Mentioned In	n/a

machine identities

Device identity material used for authenticated publish, ingest, and audit attribution.

Key	Value
Kind	Catalog Alias
Name	machine identities
Mentioned In	n/a

machine status

Machine telemetry including brew duration, temperature, and status metadata.

Key	Value
Kind	Catalog Alias
Name	machine status
Mentioned In	Architecture Intent View

machine telemetry

Machine telemetry including brew duration, temperature, and status metadata.

Key	Value
Kind	Catalog Alias
Name	machine telemetry
Mentioned In	n/a

mqtt sdk

SDK used by edge components for telemetry and command transport.

Key	Value
Kind	Catalog Alias
Name	mqtt sdk

Key	Value
Mentioned In	n/a

normal mode

Default operation mode for telemetry and OTA behavior.

Key	Value
Kind	Catalog Alias
Name	normal mode
Mentioned In	n/a

normal operation mode

Default operation mode for telemetry and OTA behavior.

Key	Value
Kind	Catalog Alias
Name	normal operation mode
Mentioned In	n/a

observability event fanout

Event raised when cloud ingestion accepts telemetry.

Key	Value
Kind	Catalog Alias
Name	observability event fanout
Mentioned In	n/a

observability events

Event raised when cloud ingestion accepts telemetry.

Key	Value
Kind	Catalog Alias
Name	observability events
Mentioned In	n/a

observability reporting

Aggregates telemetry and update outcomes into operational and audit signals.

Key	Value
Kind	Catalog Alias
Name	observability reporting
Mentioned In	n/a

ota commands

Event raised when a fleet OTA campaign is scheduled.

Key	Value
Kind	Catalog Alias
Name	ota commands
Mentioned In	n/a

ota enabled

Feature flag enabling centralized OTA rollout behavior.

Key	Value
Kind	Catalog Alias
Name	ota enabled
Mentioned In	n/a

ota outcomes

Device-level status for firmware download, apply, and rollback outcomes.

Key	Value
Kind	Catalog Alias
Name	ota outcomes
Mentioned In	Architecture Intent View

ota rollout

Feature flag enabling centralized OTA rollout behavior.

Key	Value
Kind	Catalog Alias
Name	ota rollout
Mentioned In	Architecture Intent View , REQ-COF-003 , Ota Rollout Flow

ota rollout commands

Event raised when a fleet OTA campaign is scheduled.

Key	Value
Kind	Catalog Alias
Name	ota rollout commands
Mentioned In	n/a

ota status reported

Event raised when device reports OTA apply status.

Key	Value
Kind	Catalog Alias
Name	ota status reported
Mentioned In	n/a

platform ops

Owns runtime deployment and operational controls.

Key	Value
Kind	Catalog Alias
Name	platform ops
Mentioned In	n/a

replay abuse

Event raised when replayed telemetry behavior is detected and flagged as abuse.

Key	Value
Kind	Catalog Alias
Name	replay abuse
Mentioned In	`Security View`, `REQ-COF-012`

rollback firmware apply

Device-level status for firmware download, apply, and rollback outcomes.

Key	Value
Kind	Catalog Alias

Key	Value
Name	rollback firmware apply
Mentioned In	REQ-COF-005

rollback status

Event raised when device reports OTA apply status.

Key	Value
Kind	Catalog Alias
Name	rollback status
Mentioned In	n/a

rollout cohorts

Policy describing rollout cohorts, windows, and rollback criteria.

Key	Value
Kind	Catalog Alias
Name	rollout cohorts
Mentioned In	Architecture Intent View

rollout windows

Policy describing rollout cohorts, windows, and rollback criteria.

Key	Value
Kind	Catalog Alias
Name	rollout windows
Mentioned In	n/a

runtime operations

Operates cloud runtime deployment and scaling controls.

Key	Value
Kind	Catalog Alias
Name	runtime operations
Mentioned In	n/a

security reviewer

Reviews identity, firmware integrity, and logging controls.

Key	Value
Kind	Catalog Alias
Name	security reviewer
Mentioned In	n/a

signature valid

Condition indicating firmware signature verification succeeded.

Key	Value
Kind	Catalog Alias
Name	signature valid
Mentioned In	Security View , Ota Campaign Test

signed firmware bundles

Signed OTA firmware artifact distributed to machine cohorts for update apply.

Key	Value
Kind	Catalog Alias
Name	signed firmware bundles
Mentioned In	Architecture Intent View

signer service

Service that signs firmware artifacts before OTA rollout.

Key	Value
Kind	Catalog Alias
Name	signer service
Mentioned In	n/a

status feedback

Event raised when device reports OTA apply status.

Key	Value
Kind	Catalog Alias
Name	status feedback

Key	Value
Mentioned In	n/a

store operator

Operates coffee machines and observes local machine health.

Key	Value
Kind	Catalog Alias
Name	store operator
Mentioned In	n/a

telemetry

Machine telemetry including brew duration, temperature, and status metadata.

Key	Value
Kind	Catalog Alias
Name	telemetry
Mentioned In	`Architecture Intent View`, `Deployment View`, `Security View`, `Interaction Flow View`, `REQ-COF-001`, `REQ-COF-002`, `REQ-COF-006`, `REQ-COF-008`, `REQ-COF-011`, `Ota Rollout Flow`, `Logging Pipeline Test`, `Telemetry Ingest Contract Test`

telemetry collection

Collects brew/device telemetry and publishes cloud-bound payloads.

Key	Value
Kind	Catalog Alias
Name	telemetry collection
Mentioned In	n/a

telemetry ingested

Event raised when cloud ingestion accepts telemetry.

Key	Value
Kind	Catalog Alias
Name	telemetry ingested
Mentioned In	n/a

telemetry payloads

Machine telemetry including brew duration, temperature, and status metadata.

Key	Value
Kind	Catalog Alias
Name	telemetry payloads
Mentioned In	Architecture Intent View

telemetry records

Machine telemetry including brew duration, temperature, and status metadata.

Key	Value
Kind	Catalog Alias
Name	telemetry records
Mentioned In	n/a

telemetry replay

Replayed telemetry floods ingestion and distorts reporting.

Key	Value
Kind	Catalog Alias
Name	telemetry replay
Mentioned In	n/a

telemetry replay detected

Event raised when replayed telemetry behavior is detected and flagged as abuse.

Key	Value
Kind	Catalog Alias
Name	telemetry replay detected
Mentioned In	n/a

the coffee fleet system

Connected coffee-machine fleet and central cloud control system-of-interest covered by this architecture.

Key	Value
Kind	Catalog Alias
Name	the coffee fleet system

Key	Value
Mentioned In	REQ-COF-001 , REQ-COF-002 , REQ-COF-003 , REQ-COF-004 , REQ-COF-005 , REQ-COF-006 , REQ-COF-007 , REQ-COF-008 , REQ-COF-009 , REQ-COF-010 , REQ-COF-011 , REQ-COF-012

update outcomes

Device-level status for firmware download, apply, and rollback outcomes.

Key	Value
Kind	Catalog Alias
Name	update outcomes
Mentioned In	n/a

update status

Event raised when device reports OTA apply status.

Key	Value
Kind	Catalog Alias
Name	update status
Mentioned In	n/a

update/rollback status

Device-level status for firmware download, apply, and rollback outcomes.

Key	Value
Kind	Catalog Alias
Name	update/rollback status
Mentioned In	n/a

validation failed

Event raised when signature/hash validation fails on firmware.

Key	Value
Kind	Catalog Alias
Name	validation failed
Mentioned In	n/a

AV-LOG-TAMPERING

Attempts to suppress or alter security/operational log records.

Key	Value
Kind	Catalog Attack Vector
Name	log tampering
Mentioned In	Security View

AV-MALICIOUS-FIRMWARE-BUNDLE

Tampered firmware payload attempts to execute on devices.

Key	Value
Kind	Catalog Attack Vector
Name	malicious firmware bundle
Mentioned In	Security View

AV-SPOOFED-DEVICE-IDENTITY

Unauthorized actor impersonates a machine identity.

Key	Value
Kind	Catalog Attack Vector
Name	spoofed device identity
Mentioned In	Security View

AV-TELEMETRY-REPLAY-ABUSE

Replayed telemetry floods ingestion and distorts reporting.

Key	Value
Kind	Catalog Attack Vector
Name	telemetry replay abuse
Mentioned In	Security View

COND-FIRMWARE-SIGNATURE-VALID

Condition indicating firmware signature verification succeeded.

Key	Value
Kind	Catalog Condition
Name	firmware signature is valid

Key	Value
Mentioned In	REQ-COF-004

DATA-AUDIT-LOG-RECORD

Immutable log record for security and compliance analysis.

Key	Value
Kind	Catalog Data Term
Name	audit log record
Mentioned In	Interaction Flow View , REQ-COF-007

DATA-BREW-TELEMETRY

Machine telemetry including brew duration, temperature, and status metadata.

Key	Value
Kind	Catalog Data Term
Name	brew telemetry
Mentioned In	REQ-COF-001

DATA-CAMPAIGN-ID

Stable OTA campaign identifier used to correlate rollout and audit records.

Key	Value
Kind	Catalog Data Term
Name	campaign id
Mentioned In	REQ-COF-007

DATA-FIRMWARE-BUNDLE

Signed OTA firmware artifact distributed to machine cohorts for update apply.

Key	Value
Kind	Catalog Data Term
Name	signed firmware bundle
Mentioned In	REQ-COF-003

DATA-MACHINE-IDENTITY

Device identity material used for authenticated publish, ingest, and audit attribution.

Key	Value
Kind	Catalog Data Term
Name	machine identity
Mentioned In	Interaction Flow View, REQ-COF-007

DATA-OTA-CAMPAIGN-POLICY

Policy describing rollout cohorts, windows, and rollback criteria.

Key	Value
Kind	Catalog Data Term
Name	ota campaign policy
Mentioned In	REQ-COF-003

DATA-OTA-STATUS

Device-level status for firmware download, apply, and rollback outcomes.

Key	Value
Kind	Catalog Data Term
Name	ota status
Mentioned In	REQ-COF-010

EVT-BREW-CYCLE-COMPLETED

Event raised when a coffee machine completes a brew cycle.

Key	Value
Kind	Catalog Event
Name	brew cycle completed event is received
Mentioned In	REQ-COF-001

EVT-CLOUD-INGEST-ENDPOINT-UNAVAILABLE

Event raised when cloud ingest endpoint becomes unavailable for telemetry ingestion.

Key	Value
Kind	Catalog Event
Name	cloud ingest endpoint is unavailable
Mentioned In	REQ-COF-006, REQ-COF-011

EVT-FIRMWARE-VALIDATION-FAILED

Event raised when signature/hash validation fails on firmware.

Key	Value
Kind	Catalog Event
Name	firmware validation failed event is received
Mentioned In	REQ-COF-005, REQ-COF-010

EVT-OTA-CAMPAIGN-SCHEDULED

Event raised when a fleet OTA campaign is scheduled.

Key	Value
Kind	Catalog Event
Name	ota campaign scheduled event is received
Mentioned In	REQ-COF-003, REQ-COF-004

EVT-OTA-STATUS-REPORTED

Event raised when device reports OTA apply status.

Key	Value
Kind	Catalog Event
Name	ota status reported event is received
Mentioned In	REQ-COF-007

EVT-TELEMETRY-INGESTED

Event raised when cloud ingestion accepts telemetry.

Key	Value
Kind	Catalog Event
Name	telemetry ingested event is received
Mentioned In	REQ-COF-002, REQ-COF-009

EVT-TELEMETRY-REPLAY-ABUSE-DETECTED

Event raised when replayed telemetry behavior is detected and flagged as abuse.

Key	Value
Kind	Catalog Event

Key	Value
Name	telemetry replay abuse is detected
Mentioned In	REQ-COF-008 , REQ-COF-012

FEAT-AUDIT-LOGGING

Feature flag enabling audit-grade log persistence.

Key	Value
Kind	Catalog Feature
Name	audit logging is enabled
Mentioned In	REQ-COF-007

FEAT-OTA-CAMPAIGN

Feature flag enabling centralized OTA rollout behavior.

Key	Value
Kind	Catalog Feature
Name	ota campaign is enabled
Mentioned In	REQ-COF-003 , REQ-COF-004

FG-CLOUD-CONTROL

Capability area for fleet ingestion, campaign orchestration, and cloud reporting.

Key	Value
Kind	Catalog Functional Group
Name	cloud control
Mentioned In	Architecture Intent View , Deployment View , REQ-COF-003

FG-MACHINE-EDGE

Capability area for device-local telemetry and OTA execution behavior.

Key	Value
Kind	Catalog Functional Group
Name	machine edge
Mentioned In	Architecture Intent View

FG-PLATFORM-OPERATIONS

Capability area for runtime operations and device identity/secrets lifecycle.

Key	Value
Kind	Catalog Functional Group
Name	platform operations
Mentioned In	Architecture Intent View, Deployment View

FU-CLOUD-RUNTIME-OPERATIONS

Operates cloud runtime deployment and scaling controls.

Key	Value
Kind	Catalog Functional Unit
Name	cloud runtime operations
Mentioned In	Architecture Intent View, Security View

FU-DEVICE-IDENTITY-SECRETS

Manages device identity material and secure configuration.

Key	Value
Kind	Catalog Functional Unit
Name	device identity and secrets
Mentioned In	Architecture Intent View, Communication View, Deployment View, Security View, Traceability View, Logging Pipeline Test, Ota Campaign Test

FU-FLEET-INGESTION-API

Ingests telemetry and machine status from edge devices.

Key	Value
Kind	Catalog Functional Unit
Name	fleet ingestion api
Mentioned In	Architecture Intent View, Deployment View, Security View, REQ-COF-001, Logging Pipeline Test, Telemetry Ingest Contract Test

FU-FLEET-OBSERVABILITY-REPORTING

Aggregates telemetry and update outcomes into operational and audit signals.

Key	Value
Kind	Catalog Functional Unit

Key	Value
Name	fleet observability reporting
Mentioned In	Architecture Intent View , Deployment View , Security View , Ota Rollout Flow , Logging Pipeline Test

FU-MACHINE-TELEMETRY-COLLECTION

Collects brew/device telemetry and publishes cloud-bound payloads.

Key	Value
Kind	Catalog Functional Unit
Name	machine telemetry collection
Mentioned In	Architecture Intent View , Communication View , Deployment View , Security View , Traceability View , Telemetry Ingest Contract Test

FU-OTA-UPDATE-AGENT

Downloads, validates, and applies signed firmware updates on machines.

Key	Value
Kind	Catalog Functional Unit
Name	ota update agent
Mentioned In	Architecture Intent View , Communication View , Deployment View , Security View , Traceability View , Ota Rollout Flow , Ota Campaign Test

FU-UPDATE-CAMPAIGN-ORCHESTRATION

Selects target machine groups and coordinates OTA rollout phases.

Key	Value
Kind	Catalog Functional Unit
Name	update campaign orchestration
Mentioned In	Architecture Intent View , Deployment View , Security View , Ota Rollout Flow , Ota Campaign Test

MODE-NORMAL

Default operation mode for telemetry and OTA behavior.

Key	Value
Kind	Catalog Mode
Name	normal mode is active
Mentioned In	REQ-COF-011

REF-CLOUD-LOGGING-SERVICE

Managed log and metrics service used for fleet observability.

Key	Value
Kind	Catalog Referenced Element
Name	cloud logging service
Mentioned In	Architecture Intent View , Deployment View , Security View

REF-CLOUD-RUNTIME-SDK

SDK used by cloud units for runtime integrations.

Key	Value
Kind	Catalog Referenced Element
Name	cloud runtime sdk
Mentioned In	Architecture Intent View , Deployment View , Security View

REF-FIRMWARE-BUNDLE-STORE

Object store containing signed OTA firmware bundles.

Key	Value
Kind	Catalog Referenced Element
Name	firmware bundle store
Mentioned In	Architecture Intent View , Deployment View , Security View

REF-FIRMWARE-SIGNER-SERVICE

Service that signs firmware artifacts before OTA rollout.

Key	Value
Kind	Catalog Referenced Element
Name	firmware signer service
Mentioned In	Architecture Intent View , Deployment View , Security View

REF-IOT-INGEST-ENDPOINT

Cloud endpoint for receiving telemetry and status from devices.

Key	Value
Kind	Catalog Referenced Element

Key	Value
Name	iot ingest endpoint
Mentioned In	Architecture Intent View , Deployment View , Security View

REF-MQTT-DEVICE-SDK

SDK used by edge components for telemetry and command transport.

Key	Value
Kind	Catalog Referenced Element
Name	mqtt device sdk
Mentioned In	Architecture Intent View , Deployment View , Security View

STATE-FLEET-OPERATION-ACTIVE

State where machines and central cloud are actively serving operations.

Key	Value
Kind	Catalog State
Name	coffee machine fleet operation is active
Mentioned In	REQ-COF-001 , REQ-COF-002 , REQ-COF-009

SYS-COFFEE-FLEET-SYSTEM

Connected coffee-machine fleet and central cloud control system-of-interest covered by this architecture.

Key	Value
Kind	Catalog System
Name	coffee fleet system
Mentioned In	n/a

TERM-ACTOR

A person or operational role that interacts with functional units.

Key	Value
Kind	Engineering Model Term
Name	actor
Mentioned In	n/a

TERM-ASCIIDOC-DIAGRAM

Generated diagram block, usually Mermaid, that visualizes architecture relationships in the AsciiDoc publication.

Key	Value
Kind	Engineering Model Term
Name	AsciiDoc diagram
Mentioned In	n/a

TERM-ASCIIDOC-DOCUMENT

Human-readable generated architecture publication document assembled from authored model, inferred evidence, diagrams, references, and decisions.

Key	Value
Kind	Engineering Model Term
Name	AsciiDoc document
Mentioned In	n/a

TERM-ASCIIDOC-SECTION

Structured generated document section or row model used to render authored and inferred architecture content.

Key	Value
Kind	Engineering Model Term
Name	AsciiDoc section
Mentioned In	n/a

TERM-ATTACK-VECTOR

A technical misuse or attack path that targets functional, referenced, or runtime elements.

Key	Value
Kind	Engineering Model Term
Name	attack vector
Mentioned In	n/a

TERM-AUTHORED-MAPPING

An explicit relationship declared in architecture inputs between authored or referenced elements.

Key	Value
Kind	Engineering Model Term

Key	Value
Name	authored mapping
Mentioned In	Communication View, Security View

TERM-BUNDLE

Loaded architecture, catalog, and companion documents resolved as one working model context.

Key	Value
Kind	Engineering Model Term
Name	model bundle
Mentioned In	n/a

TERM-CATALOG

Controlled vocabulary document used to normalize model, requirement, and generated-document terminology.

Key	Value
Kind	Engineering Model Term
Name	catalog
Mentioned In	n/a

TERM-CATALOG-ENTRY

A stable catalog term with a definition and aliases for linting, linking, and generated references.

Key	Value
Kind	Engineering Model Term
Name	catalog entry
Mentioned In	n/a

TERM-CLI-COMMAND

Command-line entrypoint that coordinates loading, validation, generation, and file output for an engineering-model workflow.

Key	Value
Kind	Engineering Model Term
Name	CLI command
Mentioned In	n/a

TERM-CODE-ELEMENT

An inferred code structure or ownership element discovered from source trees and build metadata.

Key	Value
Kind	Engineering Model Term
Name	code element
Mentioned In	n/a

TERM-COMPLIANCE-MAPPING

Authored mapping that links a model control to selected compliance controls, model scope, implementation status, evidence, and responsible roles.

Key	Value
Kind	Engineering Model Term
Name	compliance mapping
Mentioned In	n/a

TERM-CONTROL

An authored security, policy, or operational control used to constrain behavior and reduce risk.

Key	Value
Kind	Engineering Model Term
Name	control
Mentioned In	Architecture Intent View, Deployment View, REQ-COF-004, REQ-COF-008

TERM-CONTROL-VERIFICATION

Authored control verification record with method, status, threat/risk scope, findings, and evidence.

Key	Value
Kind	Engineering Model Term
Name	control verification
Mentioned In	n/a

TERM-DATA-OBJECT

An authored data artifact or contract shape traced across flow, interface, and implementation boundaries.

Key	Value
Kind	Engineering Model Term

Key	Value
Name	data object
Mentioned In	n/a

TERM-DECISION

Authored architecture decision record with status, context, decision text, and consequences.

Key	Value
Kind	Engineering Model Term
Name	architecture decision
Mentioned In	n/a

TERM-DEPLOYMENT-TARGET

An authored deployment destination such as environment, cluster, namespace, or registry scope.

Key	Value
Kind	Engineering Model Term
Name	deployment target
Mentioned In	n/a

TERM-DESIGN

Authored design narrative organized by model entities and architecture views.

Key	Value
Kind	Engineering Model Term
Name	design document
Mentioned In	n/a

TERM-DESIGN-VIEW

View-scoped design narrative attached to an authored functional group or functional unit.

Key	Value
Kind	Engineering Model Term
Name	design view
Mentioned In	n/a

TERM-EVENT

An authored trigger signal that drives transitions, flow progress, or asynchronous outcomes.

Key	Value
Kind	Engineering Model Term
Name	event
Mentioned In	Communication View , Security View , State Lifecycle View , REQ-COF-012

TERM-EVIDENCE

A file, artifact, or observation used to support control, risk, threat, or verification claims.

Key	Value
Kind	Engineering Model Term
Name	evidence
Mentioned In	Communication View , Deployment View , Security View , Traceability View , State Lifecycle View , Interaction Flow View

TERM-FLOW

An authored causal interaction sequence from user/system intent to outcome, represented as ordered steps.

Key	Value
Kind	Engineering Model Term
Name	flow
Mentioned In	Architecture Intent View , Communication View , Interaction Flow View

TERM-FLOW-STEP

A single authored step in a flow that captures action, data in/out, references, and normal/error/async transitions.

Key	Value
Kind	Engineering Model Term
Name	flow step
Mentioned In	n/a

TERM-FUNCTIONAL-GROUP

A major authored capability area that groups related functional units.

Key	Value
Kind	Engineering Model Term

Key	Value
Name	functional group
Mentioned In	n/a

TERM-FUNCTIONAL-UNIT

An authored working unit inside a functional group that owns specific behavior.

Key	Value
Kind	Engineering Model Term
Name	functional unit
Mentioned In	`State Lifecycle View`

TERM-GENERATION-WORKFLOW

Orchestrated run that combines model loading, validation, projection, rendering, and artifact emission.

Key	Value
Kind	Engineering Model Term
Name	generation workflow
Mentioned In	n/a

TERM-INFERENCE-HINT

Authored source and ownership configuration used to discover runtime, code, and verification evidence.

Key	Value
Kind	Engineering Model Term
Name	inference hint
Mentioned In	n/a

TERM-INFERRED-MAPPING

A discovered relationship that links inferred runtime/code elements upward to authored design.

Key	Value
Kind	Engineering Model Term
Name	inferred mapping
Mentioned In	n/a

TERM-INTERFACE

An authored interface boundary (for example API, channel, endpoint, or contract) owned by a functional unit.

Key	Value
Kind	Engineering Model Term
Name	interface
Mentioned In	Communication View

TERM-LINT-RUN

Requirement lint configuration that controls parsing and quality checks for requirement text.

Key	Value
Kind	Engineering Model Term
Name	lint run
Mentioned In	n/a

TERM-LOBSTER-ACTIVITY-TRACE

Generated LOBSTER activity JSON that links verification evidence back to formal requirement identifiers.

Key	Value
Kind	Engineering Model Term
Name	LOBSTER activity trace
Mentioned In	n/a

TERM-MCP-SERVER

Model Context Protocol server that exposes engineering-model operations to AI agents through JSON-RPC tools.

Key	Value
Kind	Engineering Model Term
Name	MCP server
Mentioned In	n/a

TERM-MCP-STDIO-TRANSPORT

Content-Length framed standard input/output transport used by the MCP command-line server.

Key	Value
Kind	Engineering Model Term
Name	MCP stdio transport

Key	Value
Mentioned In	n/a

TERM-MCP-TOOL

Named MCP operation with a constrained input schema and structured response payload.

Key	Value
Kind	Engineering Model Term
Name	MCP tool
Mentioned In	n/a

TERM-MCP-TOOL-RESPONSE

Structured MCP tool payload that includes success state, schema version, generated timestamp, and result data or validation error.

Key	Value
Kind	Engineering Model Term
Name	MCP tool response
Mentioned In	n/a

TERM-MODEL

The authored architecture model root that composes functional structure, relationships, inference hints, and views.

Key	Value
Kind	Engineering Model Term
Name	model
Mentioned In	Security View

TERM-OPEN-OTM-DOCUMENT

Open Threat Model JSON representation generated from the engineering model for interoperable threat-model exchange.

Key	Value
Kind	Engineering Model Term
Name	Open OTM document
Mentioned In	n/a

TERM-OSCAL-ASSESSMENT-RESULTS

Generated OSCAL assessment results that summarize reviewed controls, verification findings, and modeled risks.

Key	Value
Kind	Engineering Model Term
Name	OSCAL assessment results
Mentioned In	n/a

TERM-OSCAL-POAM

Generated OSCAL plan of action and milestones that maps modeled POA&M items and related risks into compliance JSON.

Key	Value
Kind	Engineering Model Term
Name	OSCAL POA&M
Mentioned In	n/a

TERM-OSCAL-SSP

Generated OSCAL system security plan that maps authored system metadata, components, controls, allocations, and evidence into SSP JSON.

Key	Value
Kind	Engineering Model Term
Name	OSCAL SSP
Mentioned In	n/a

TERM-POAM-ITEM

Plan of action and milestones item linked to a modeled risk and supporting artifacts.

Key	Value
Kind	Engineering Model Term
Name	POA&M item
Mentioned In	n/a

TERM-REFERENCE-INDEX

Generated index of authored, catalog, runtime, code, and verification references with stable anchors and backlinks.

Key	Value
Kind	Engineering Model Term
Name	reference index

Key	Value
Mentioned In	n/a

TERM-REFERENCED-ELEMENT

An architecture-relevant external, platform, or third-party dependency represented by role.

Key	Value
Kind	Engineering Model Term
Name	referenced element
Mentioned In	n/a

TERM-REPO-INDEX

Bounded first-party source index used to answer model-aware file, requirement, control, and threat lookup queries.

Key	Value
Kind	Engineering Model Term
Name	repository index
Mentioned In	n/a

TERM-REQUIREMENT

A structured requirement statement that defines expected system behavior and maps to functional ownership.

Key	Value
Kind	Engineering Model Term
Name	requirement
Mentioned In	Traceability View , Interaction Flow View , REQ-COF-002 , REQ-COF-007 , REQ-COF-008 , REQ-COF-009 , REQ-COF-010 , REQ-COF-012

TERM-RISK

Authored risk record with likelihood, impact, response, scope, controls, and evidence.

Key	Value
Kind	Engineering Model Term
Name	risk
Mentioned In	Deployment View , Security View

TERM-RUNTIME-ELEMENT

An inferred runtime realization element discovered from infrastructure and deployment sources.

Key	Value
Kind	Engineering Model Term
Name	runtime element
Mentioned In	n/a

TERM-SECURITY-CONTEXT

Security-focused generated context view that groups owned functional units and shows external actors, references, flows, controls, and trust boundaries.

Key	Value
Kind	Engineering Model Term
Name	security context
Mentioned In	n/a

TERM-SOURCE-BLOCK

Stable source reference block that records file path, optional line span, kind, summary, and linked entity IDs.

Key	Value
Kind	Engineering Model Term
Name	source block
Mentioned In	n/a

TERM-STATE

An authored lifecycle state used to model transition behavior, guards, and event-driven progression.

Key	Value
Kind	Engineering Model Term
Name	state
Mentioned In	`State Lifecycle View`

TERM-STRUCTURIZR-DSL

Generated Structurizr DSL workspace that projects authored architecture, relationships, dynamic views, and deployment metadata.

Key	Value
Kind	Engineering Model Term

Key	Value
Name	Structurizr DSL
Mentioned In	n/a

TERM-THREAT-ASSUMPTION

Authored security assumption that records scope, rationale, owner, status, and evidence.

Key	Value
Kind	Engineering Model Term
Name	threat assumption
Mentioned In	n/a

TERM-THREAT-DRAGON-DOCUMENT

Threat Dragon JSON representation generated from the engineering model for STRIDE threat-model review.

Key	Value
Kind	Engineering Model Term
Name	Threat Dragon document
Mentioned In	n/a

TERM-THREAT-MITIGATION

Authored mitigation record that links a threat scenario to a control, effectiveness, verification, and evidence.

Key	Value
Kind	Engineering Model Term
Name	threat mitigation
Mentioned In	n/a

TERM-THREAT-MODEL-EXPORT

Generated security artifact that translates authored architecture, flows, trust boundaries, threats, and mitigations into an external threat-model schema.

Key	Value
Kind	Engineering Model Term
Name	threat model export
Mentioned In	n/a

TERM-THREAT-OUT-OF-SCOPE

Authored decision that excludes a threat concern from current scope with reason, owner, expiry, and evidence.

Key	Value
Kind	Engineering Model Term
Name	threat out-of-scope decision
Mentioned In	n/a

TERM-THREAT-SCENARIO

Authored threat narrative that connects attack vectors, scope, flows, controls, risks, and verification evidence.

Key	Value
Kind	Engineering Model Term
Name	threat scenario
Mentioned In	n/a

TERM-TRACE-MARKER

Source-level marker such as TRLC-LINKS or ENGMODEL-LINKS used to connect declarations to requirements and model entities.

Key	Value
Kind	Engineering Model Term
Name	trace marker
Mentioned In	n/a

TERM-TRLC-PACKAGE

Generated TRLC model and requirements package used for formal requirement traceability processing.

Key	Value
Kind	Engineering Model Term
Name	TRLC package
Mentioned In	n/a

TERM-TRUST-BOUNDARY

An authored trust separation zone that marks policy, identity, or security control boundaries.

Key	Value
Kind	Engineering Model Term

Key	Value
Name	trust boundary
Mentioned In	Architecture Intent View , Security View , REQ-COF-004

TERM-UPWARD-LINKING

Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs.

Key	Value
Kind	Engineering Model Term
Name	upward linking
Mentioned In	n/a

TERM-VALIDATION

Model quality gate that checks authored documents, references, relationship semantics, and requirement lint results.

Key	Value
Kind	Engineering Model Term
Name	validation
Mentioned In	Architecture Intent View

TERM-VALIDATION-DIAGNOSTIC

Structured validation finding with code, severity, message, and optional source path.

Key	Value
Kind	Engineering Model Term
Name	validation diagnostic
Mentioned In	n/a

TERM-VERIFICATION-CHECK

An inferred or authored verification artifact that validates requirement behavior with test evidence.

Key	Value
Kind	Engineering Model Term
Name	verification check
Mentioned In	n/a

TERM-VIEW

Authored projection configuration that selects roots, entity kinds, mappings, audience, and abstraction.

Key	Value
Kind	Engineering Model Term
Name	view
Mentioned In	Communication View , Deployment View , Security View , Traceability View , State Lifecycle View , Interaction Flow View

14.3. Inferred Runtime References

ota_campaign_schedule

schedules OTA campaign orchestration triggers for configured rollout windows

Key	Value
Kind	Runtime eventbridge_rule
Owner	FU-CLOUD-RUNTIME-OPERATIONS
Source	examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf
Mentioned In	Deployment View

ota_campaign_target

routes scheduled OTA campaign events to the orchestration runtime handler

Key	Value
Kind	Runtime eventbridge_target
Owner	FU-CLOUD-RUNTIME-OPERATIONS
Source	examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf
Mentioned In	Deployment View

fleet_ingestion_api

receives device telemetry, validates payload shape, and publishes normalized fleet events

Key	Value
Kind	Runtime lambda_function
Owner	FU-FLEET-INGESTION-API
Source	examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf
Mentioned In	Deployment View , Security View

fleet_observability_reporting

aggregates telemetry outcomes into fleet operations and security reporting signals

Key	Value
Kind	Runtime lambda_function
Owner	FU-FLEET-OBSERVABILITY-REPORTING
Source	examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf
Mentioned In	Deployment View, Security View

update_campaign_orchestration

coordinates OTA campaign rollout windows and device cohort targeting

Key	Value
Kind	Runtime lambda_function
Owner	FU-UPDATE-CAMPAIGN-ORCHESTRATION
Source	examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf
Mentioned In	Deployment View, Security View

telemetry_retry_queue

buffers telemetry during ingest outages for retry once normal mode resumes

Key	Value
Kind	Runtime queue
Owner	FU-CLOUD-RUNTIME-OPERATIONS
Source	examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf
Mentioned In	Deployment View

fleet_event_topic

broadcasts fleet-wide telemetry and OTA lifecycle events to subscribed processors

Key	Value
Kind	Runtime topic
Owner	FU-CLOUD-RUNTIME-OPERATIONS
Source	examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf
Mentioned In	Deployment View

14.4. Inferred Code References

cloud_runtime_ops.go

wires cloud runtime handlers and integration entrypoints

Key	Value
Kind	Code source_file
Owner	FU-CLOUD-RUNTIME-OPERATIONS
Source	cloud_runtime_ops.go
Mentioned In	Deployment View

device_identity_secrets.ts

resolves device identity credentials and firmware verification secrets

Key	Value
Kind	Code source_file
Owner	FU-DEVICE-IDENTITY-SECRETS
Source	device_identity_secrets.ts
Mentioned In	Deployment View , Security View

fleet_ingestion_api.go

ingests device telemetry and emits normalized fleet ingestion events

Key	Value
Kind	Code source_file
Owner	FU-FLEET-INGESTION-API
Source	fleet_ingestion_api.go
Mentioned In	n/a

fleet_observability_reporting.ts

aggregates telemetry and OTA outcome signals for fleet observability reporting

Key	Value
Kind	Code source_file
Owner	FU-FLEET-OBSERVABILITY-REPORTING
Source	fleet_observability_reporting.ts

Key	Value
Mentioned In	n/a

machine_telemetry.rs

builds machine telemetry and status payloads at the edge

Key	Value
Kind	Code source_file
Owner	FU-MACHINE-TELEMETRY-COLLECTION
Source	machine_telemetry.rs
Mentioned In	Deployment View , Security View

ota_update_agent.rs

applies OTA campaign updates and reports rollout outcomes

Key	Value
Kind	Code source_file
Owner	FU-OTA-UPDATE-AGENT
Source	ota_update_agent.rs
Mentioned In	Deployment View , Security View

update_campaign_orchestrator.ts

plans OTA rollout cohorts and campaign lifecycle transitions

Key	Value
Kind	Code source_file
Owner	FU-UPDATE-CAMPAIGN-ORCHESTRATION
Source	update_campaign_orchestrator.ts
Mentioned In	Deployment View , Security View

cloud_runtime_ops.go:6

Inferred code symbol owned by FU-CLOUD-RUNTIME-OPERATIONS from cloud_runtime_ops.go:6.

Key	Value
Kind	Code symbol
Owner	FU-CLOUD-RUNTIME-OPERATIONS

Key	Value
Source	cloud_runtime_ops.go:6
Mentioned In	n/a

device_identity_secrets.ts:10

Inferred code symbol owned by FU-DEVICE-IDENTITY-SECRETS from device_identity_secrets.ts:10.

Key	Value
Kind	Code symbol
Owner	FU-DEVICE-IDENTITY-SECRETS
Source	device_identity_secrets.ts:10
Mentioned In	n/a

device_identity_secrets.ts:5

Inferred code symbol owned by FU-DEVICE-IDENTITY-SECRETS from device_identity_secrets.ts:5.

Key	Value
Kind	Code symbol
Owner	FU-DEVICE-IDENTITY-SECRETS
Source	device_identity_secrets.ts:5
Mentioned In	n/a

fleet_ingestion_api.go:13

Inferred code symbol owned by FU-FLEET-INGESTION-API from fleet_ingestion_api.go:13.

Key	Value
Kind	Code symbol
Owner	FU-FLEET-INGESTION-API
Source	fleet_ingestion_api.go:13
Mentioned In	n/a

fleet_ingestion_api.go:19

Inferred code symbol owned by FU-FLEET-INGESTION-API from fleet_ingestion_api.go:19.

Key	Value
Kind	Code symbol

Key	Value
Owner	FU-FLEET-INGESTION-API
Source	fleet_ingestion_api.go:19
Mentioned In	n/a

fleet_ingestion_api.go:6

Inferred code symbol owned by FU-FLEET-INGESTION-API from fleet_ingestion_api.go:6.

Key	Value
Kind	Code symbol
Owner	FU-FLEET-INGESTION-API
Source	fleet_ingestion_api.go:6
Mentioned In	n/a

fleet_observability_reporting.ts:12

Inferred code symbol owned by FU-FLEET-OBSERVABILITY-REPORTING from fleet_observability_reporting.ts:12.

Key	Value
Kind	Code symbol
Owner	FU-FLEET-OBSERVABILITY-REPORTING
Source	fleet_observability_reporting.ts:12
Mentioned In	n/a

fleet_observability_reporting.ts:18

Inferred code symbol owned by FU-FLEET-OBSERVABILITY-REPORTING from fleet_observability_reporting.ts:18.

Key	Value
Kind	Code symbol
Owner	FU-FLEET-OBSERVABILITY-REPORTING
Source	fleet_observability_reporting.ts:18
Mentioned In	n/a

fleet_observability_reporting.ts:6

Inferred code symbol owned by FU-FLEET-OBSERVABILITY-REPORTING from fleet_observability_reporting.ts:6.

Key	Value
Kind	Code symbol
Owner	FU-FLEET-OBSERVABILITY-REPORTING
Source	fleet_observability_reporting.ts:6
Mentioned In	n/a

machine_telemetry.rs:12

Inferred code symbol owned by FU-MACHINE-TELEMETRY-COLLECTION from machine_telemetry.rs:12.

Key	Value
Kind	Code symbol
Owner	FU-MACHINE-TELEMETRY-COLLECTION
Source	machine_telemetry.rs:12
Mentioned In	n/a

machine_telemetry.rs:18

Inferred code symbol owned by FU-MACHINE-TELEMETRY-COLLECTION from machine_telemetry.rs:18.

Key	Value
Kind	Code symbol
Owner	FU-MACHINE-TELEMETRY-COLLECTION
Source	machine_telemetry.rs:18
Mentioned In	n/a

machine_telemetry.rs:5

Inferred code symbol owned by FU-MACHINE-TELEMETRY-COLLECTION from machine_telemetry.rs:5.

Key	Value
Kind	Code symbol
Owner	FU-MACHINE-TELEMETRY-COLLECTION
Source	machine_telemetry.rs:5
Mentioned In	n/a

ota_update_agent.rs:16

Inferred code symbol owned by FU-OTA-UPDATE-AGENT from ota_update_agent.rs:16.

Key	Value
Kind	Code symbol
Owner	FU-OTA-UPDATE-AGENT
Source	ota_update_agent.rs:16
Mentioned In	n/a

ota_update_agent.rs:6

Inferred code symbol owned by FU-OTA-UPDATE-AGENT from ota_update_agent.rs:6.

Key	Value
Kind	Code symbol
Owner	FU-OTA-UPDATE-AGENT
Source	ota_update_agent.rs:6
Mentioned In	n/a

update_campaign_orchestrator.ts:12

Inferred code symbol owned by FU-UPDATE-CAMPAIGN-ORCHESTRATION from update_campaign_orchestrator.ts:12.

Key	Value
Kind	Code symbol
Owner	FU-UPDATE-CAMPAIGN-ORCHESTRATION
Source	update_campaign_orchestrator.ts:12
Mentioned In	n/a

update_campaign_orchestrator.ts:6

Inferred code symbol owned by FU-UPDATE-CAMPAIGN-ORCHESTRATION from update_campaign_orchestrator.ts:6.

Key	Value
Kind	Code symbol
Owner	FU-UPDATE-CAMPAIGN-ORCHESTRATION
Source	update_campaign_orchestrator.ts:6
Mentioned In	n/a

14.5. Verification References

VER-INF-E2E-OTA_ROLLOUT_FLOW-8CBDAC

runs fleet OTA rollout scenario including scheduling, deployment, and telemetry confirmation

Key	Value
Kind	e2e
Status	not-run
Source	tests/e2e/ota_rollout_flow.yaml
Name	Ota Rollout Flow
Mentioned In	Interaction Flow View , Ota Rollout Flow

VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-B39BBF

validates telemetry logging pipeline persistence and operational reporting signals

Key	Value
Kind	integration
Status	partial
Source	test-results/integration-logging.json (+1)
Name	Logging Pipeline Test
Mentioned In	Interaction Flow View , Logging Pipeline Test

VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-394794

validates telemetry ingest contract fields and accepted request semantics

Key	Value
Kind	test
Status	pass
Source	test-results/contract-telemetry.json (+1)
Name	Telemetry Ingest Contract Test
Mentioned In	Interaction Flow View , Telemetry Ingest Contract Test

VER-INF-UNIT-OTA_CAMPAIGN_TEST-5CC0D8

verifies OTA campaign signature validation and rollout trigger behavior

Key	Value
Kind	unit
Status	pass

Key	Value
Source	test-results/unit-ota.xml (+1)
Name	Ota Campaign Test
Mentioned In	Interaction Flow View, Ota Campaign Test